



slington college
(इस्लिङ्टन कलेज)

Module Code & Module Title

**CS6P05NI Project
Interim Report**

25% Individual Coursework

Submission: Final Submission

AY 2025 2026

Academic Semester: Autumn Semester 2025

Credit: 30 Credit Year Long Module

Project Title: Resilio-Wellness App

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Assignment Due Date: Tuesday, December 30, 2025

Assignment Submission Date: Tuesday, December 30, 2025

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I confirm that I understand my coursework needs to be submitted online via MST Classroom under the relevant module page before the deadline for my assignment to be accepted and marked. I am fully aware that late submissions will be treated as non-submission and a mark of zero will be awarded.

Acknowledgement

I express my thanks to my external supervisor, Ms. Oshirya Manandhar, for her consistent guidance and support during the preparation of this proposal. Her direction helped me shape the idea and move the project in the right path.

I also thank my internal supervisor, Mr. Aadesh Tandukar, for reviewing my work and offering helpful suggestions. His feedback strengthened the project ideas and improved its clarity.

I am grateful to Islington College and London Metropolitan University for providing the academic environment and resources needed to begin this project. Their support encouraged me to develop this project with confidence.

Abstract

This interim report documents the development progress of Resilio, a mobile application designed to support mental health and emotional wellness in Nepal. The app targets users experiencing stress, anxiety, low mood, or ongoing emotional pressure who cannot access traditional mental health services due to cost, distance, or stigma. Unlike existing solutions that fragment care across multiple apps or ignore local context, Resilio integrates therapist booking, live video sessions, mood tracking, questionnaires, wellness games, and motivational content within a single offline-capable platform.

The project follows Rational Unified Process methodology with defined milestones across four phases. Inception and Elaboration phases completed on schedule, delivering project vision, stakeholder requirements, system architecture, UI/UX designs, and working prototypes validating WebRTC video calling, offline synchronization, and eSewa payment integration. Construction began January 2025 with authentication, onboarding, and navigation structure now functional. Overall progress stands at 38 percent.

The primary objectives are to help users become more aware of their emotional patterns, connect with licensed therapists through affordable video sessions, and access self-help tools that work without constant internet. Remaining development covers content systems, offline support, therapist booking, wellness games, payments, and admin features. The project remains on track for completion by April 30, 2025, delivering a functional product built specifically for Nepal's mental health needs.

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1. Introduction

1.1. Introduction to Topic

Mental health touches nearly every family, but most people still struggle to talk about it. Depression affects over 280 million people worldwide and ranks among the top causes of disability (World Health Organization, 2023). The real tragedy is not that mental disorders exist. It is that help exists too, but most people never get it. In wealthy countries, about half of those with mental health problems receive treatment. In poor countries, fewer than one in ten do (Patel et al., 2018). Therapy works. The systems to deliver it just are not there.

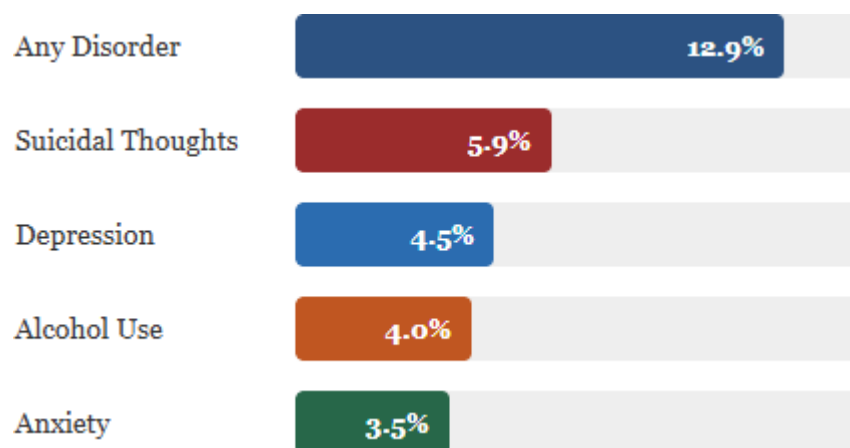


Figure 1: Prevalence of Mental Disorders in Nepal (Source: Ministry of Health and Population)

Nepal illustrates this perfectly. The Nepal Mental Health Survey found that 12.9 percent of adults have experienced mental health conditions, mostly depression, anxiety, and alcohol problems (Ministry of Health and Population, 2020). The country has roughly 100 psychiatrists for 30 million people. Most practice in Kathmandu. Someone in Jumla might travel three days just to see one. Then there is stigma. (Kohrt and Harper, 2008) documented how many Nepalis blame karma or spirits rather than seeing mental distress as medical. Seeking help feels like admitting something shameful, so people stay quiet.

But Nepal has an odd advantage. Mobile phones spread faster than anyone expected. Over 40 million subscriptions exist for 30 million people (Nepal Telecommunications Authority, 2022). Even remote villages have smartphones. Research shows digital tools can reach where traditional services cannot. Video therapy works as well as face-to-face sessions (Backhaus et al., 2012). Apps provide real support when designed properly (Naslund et al., 2017). The missing piece is an app built for Nepal. International options cost \$60-90 yearly, need credit cards and constant internet, and work only in English (Chandrashekar, 2018). Local apps offer basic tips with no therapist connection. None work offline. Resilio is being developed to solve exactly these problems, bringing therapist booking, video sessions, mood tracking, games, and motivational content into one offline-capable platform designed for Nepal's reality.

1.2. Current Scenario

Mental health conditions affect 12.9 percent of Nepali adults according to the Nepal Mental Health Survey (Ministry of Health and Population, 2020). Depression, anxiety, and alcohol use disorder are most common. Around 6 percent have experienced suicidal thoughts, and 1.4 percent have attempted suicide. These numbers translate to roughly 1.7 million adults who have considered ending their lives.

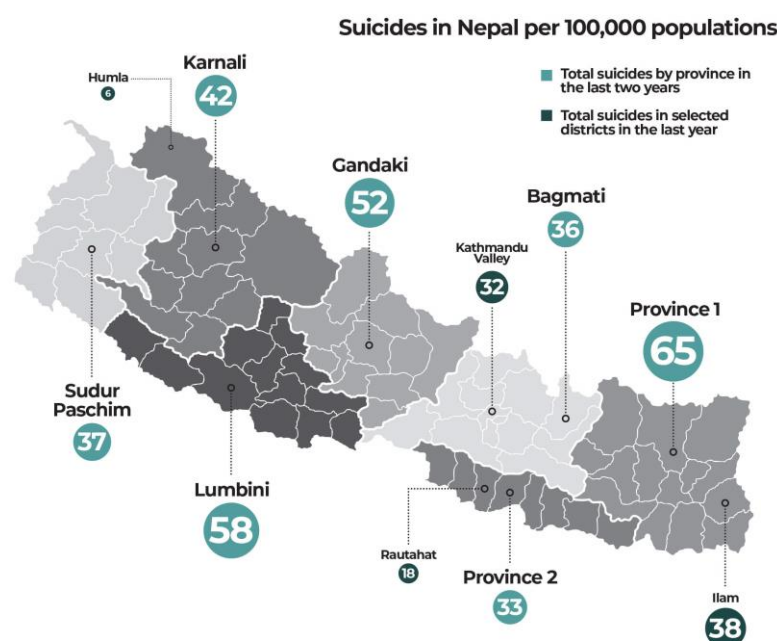


Figure 2: Suicide and Mental Health Statistics in Nepal (Source: Ministry of Health and Population, 2020)

Yet over 90 percent never receive professional care (Luitel et al., 2015). The country has roughly 100 psychiatrists for 30 million people, which works out to 0.36 per 100,000, far below WHO's recommendation of 1.0 (World Health Organization, 2017). Most practice in Kathmandu. Someone in Humla might travel three days just to see one. Cost, distance, and stigma create additional barriers. (Kohrt and Harper, 2008) documented how many Nepalis understand mental distress through spiritual frameworks rather than medical ones. Seeking help means admitting something shameful.

COVID-19 made everything urgent. WHO reported a 25 percent global increase in anxiety and depression during 2020 (World Health Organization, 2022). Nepal saw similar patterns with helplines reporting surges (Sigdel et al., 2021). But research confirmed digital solutions work. Video therapy achieves outcomes comparable to face-to-face sessions (Backhaus et al., 2012). Apps provide meaningful support in low-resource settings (Naslund et al., 2017). Nepal has an advantage here. Over 40 million mobile subscriptions exist for 30 million people (Nepal Telecommunications Authority, 2022). Smartphones reach even remote villages.

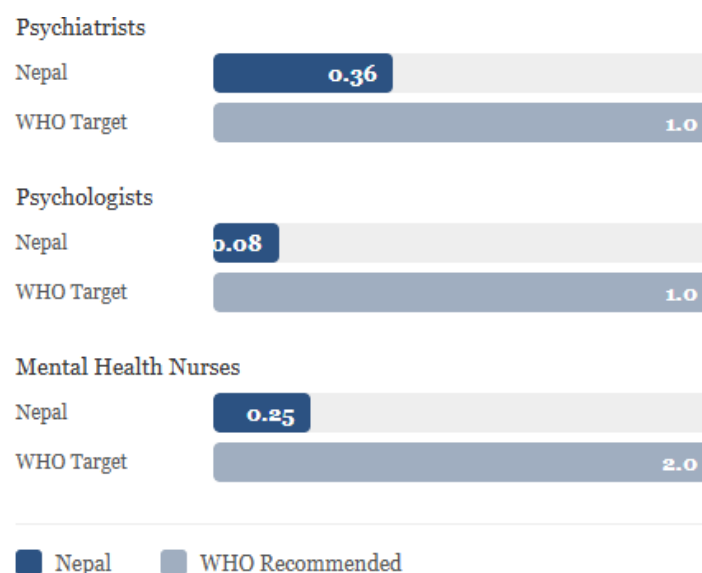


Figure 3: Mental Health Workforce per 100,000 Population (Source: World Health Organization)

[For detailed statistics please see [Appendix A: Detailed Mental Health Statistics in Nepal](#)]

1.3. Problem Statement

The core problem is not that mental health apps do not exist. They do. The problem is that none work properly for Nepal. Existing apps fail in three key ways:

- **Cost and Accessibility Barriers:**

International apps like Calm and Headspace charge around \$70 yearly. BetterHelp costs \$60 to \$90 weekly. For many Nepali families, that exceeds their entire annual healthcare spending. These apps require credit cards, stable internet, and operate only in English. They were designed for New York or London, not Dolakha or Doti (Chandrashekar, 2018).

Feature	BetterHelp	Calm	Headspace	Local Apps	Resilio
Cost & Accessibility					
Monthly Cost	\$60-90	\$15	\$13	Free-\$5	Free-\$10
Local Payment (eSewa)	No	No	No	Some	Yes
Offline Mode	No	Partial	Partial	No	Yes
Nepali Language	No	No	No	Yes	Yes
Professional Support					
Licensed Therapists	Yes	No	No	No	Yes
Live Sessions	Yes	No	No	No	Yes
Local Therapists	No	-	-	No	Yes
Self-Help Features					
Mood Tracking	Basic	Yes	Yes	Basic	Yes
Guided Activities	No	Yes	Yes	Limited	Yes
Gamified Exercises	No	Some	Some	No	Yes
Assessment Tools	Basic	No	No	No	Yes

Figure 4: Comparison of Mental Health Apps Available in Nepal

- **Local Apps Lack Depth:**

A few Nepali apps offer breathing exercises but none connect users to actual therapists. Most copy Western interfaces without cultural adaptation. Worse, none function offline. With 30 percent of Nepalis lacking regular internet (Nepal Telecommunications Authority, 2022), apps requiring constant connectivity exclude exactly those who need help most.

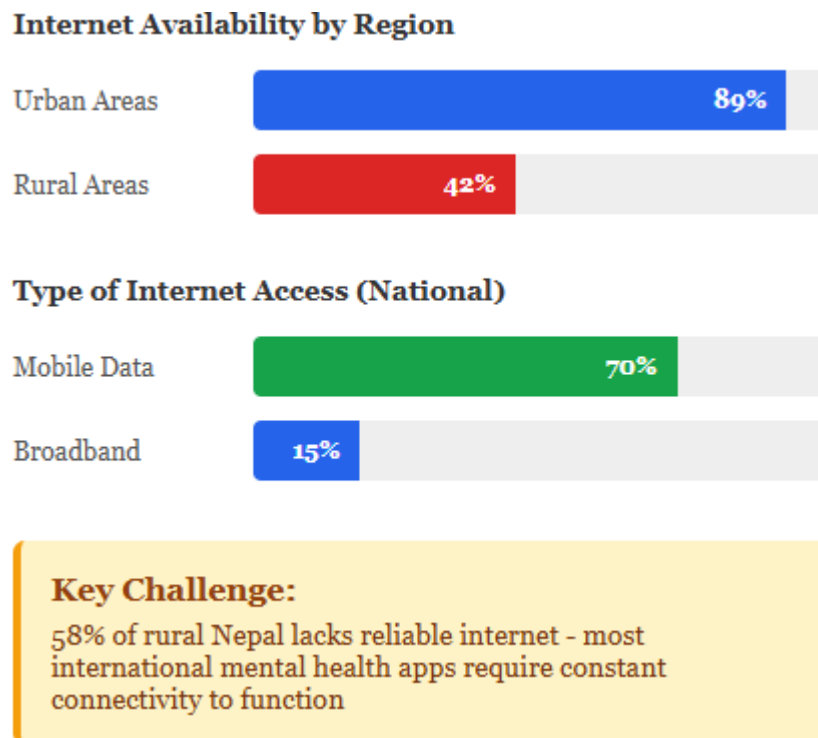


Figure 5: Internet Access Challenges for Digital Mental Health in Nepal (Source: Nepal Telecommunications Authority)

- **Fragmentation Problem:**

Users download separate apps for meditation, mood tracking, and therapy. These tools do not talk to each other. (Bakker et al., 2016) found people download an average of 3.2 apps but 74 percent abandon them within two weeks, feeling disconnected from real care. (Torous et al., 2019) reviewed over 1,000 mental health apps and found:

- Only 5 percent offered therapist integration
- Only 12 percent worked offline
- Only 3 percent had evidence-based content

The result: Nepalis find expensive foreign apps, basic local apps, or fragmented tools that never connect to real care. Most give up. The 92 percent treatment gap persists (Luitel et al., 2015).

[For extended problem analysis please see [Appendix B: Extended Problem Analysis](#)]

1.4. Project as a Solution

Resilio addresses each gap that existing apps leave open. It is not another meditation app. It is not another therapy-only platform. It is an integrated system that brings everything together in one place, works without constant internet, and fits Nepali realities instead of ignoring them.



Figure 6: Resilio - Wellness and Mental Health App Logo

- **Integration Instead of Fragmentation**

Therapist booking, live video sessions, mood tracking, questionnaires, wellness games, and motivational content exist within a single application. When users log their mood, therapists see that data before sessions. Questionnaire responses inform therapist matching. Self-help exercises link to clinical guidance. (Bakker et al., 2016) identified disconnection from real care as why people abandon mental health apps. Resilio makes everything work together.

- **Offline-First Architecture**

SQLite stores data locally. LRU caching keeps content accessible without internet. Users in villages with unreliable signal can log mood, complete exercises, and access content. Everything syncs automatically when connectivity returns. This matters because 30 percent of Nepalis lack regular internet (Nepal Telecommunications Authority, 2022).

- **Nepali Language and Culture**

Full Nepali language support with content drawn from local literature and wisdom. Therapists speak Nepali and understand local context. Discreet notifications and interface address stigma directly. (Naeem et al., 2021) confirmed cultural adaptation significantly improves outcomes in South Asian settings.

- **Gamification for Engagement**

Breathing games, wellness trivia, affirmation builders, and role-playing journeys keep users engaged. (Fleming et al., 2019) found gamification increases retention by 40 to 60 percent.

- **Accessible Payment**

ESewa integration removes credit card barriers. Free tier provides self-help tools. Paid subscriptions are priced for Nepali incomes, not Silicon Valley assumptions. (Luitel et al., 2015) identified cost as the top barrier to treatment.

The goal is not replacing traditional services. Nepal needs more psychiatrists. But those take years to train. Resilio extends limited professionals through technology, offering support for people who need help today.

[For detailed feature breakdown please see [Appendix D: Detailed Solution Features](#)]

1.5. Aim and Objectives

1.5.1. Aim

The main aim of this project is to develop Resilio, a cross-platform wellness application providing personalized mental health support, interactive wellness content, and secure access to licensed therapists, following user-centred design principles and evidence-based intervention strategies.

1.5.2. Objectives

- Develop a cross-platform mobile application using Flutter and Dart with Node.js backend ensuring code maintainability.
- Implement offline functionality using SQLite, LRU caching, and Protocol Buffer serialization for uninterrupted access.
- Create mood tracking and lifestyle logging features based on ecological momentary assessment principles.
- Develop multimedia players supporting audio, video, and images aligned with relaxation response research.
- Build interactive wellness games including breathing exercises, trivia, affirmation builders, and mental health role-playing games leveraging gamification.
- Implement therapist booking with questionnaire-based recommendations and secure WebRTC-based video/audio sessions.
- Integrate Esewa payment gateway following secure transaction protocols.
- Create administrative dashboards for content, user, and therapist management.
- Ensure compliance with data protection standards and accessibility guidelines.

1.6. Report Structure

This interim report documents Resilio's development through the Inception and Elaboration phases of the Rational Unified Process. The report covers background research, work completed so far, progress analysis, and remaining tasks.

1.6.1. Background

The background section identifies who will use Resilio: primary users seeking mental health support, licensed therapists providing services, and administrators managing the platform. It explains what the project involves, covering the six core functional areas of authentication, therapist booking, live sessions, mood tracking, questionnaires, and wellness content. Three similar applications are reviewed: BetterHelp for online therapy, Headspace for meditation, and Wysa for AI-powered support. A comparison table and analysis show how Resilio addresses gaps these apps leave open in terms of cost, offline capability, language, and integration.

1.6.2. Development

The development section covers methodology selection. Three SDLC models were evaluated: Agile, Prototype, and Rational Unified Process. RUP was selected for its balance of iteration with structured documentation and alignment with academic milestones. The work breakdown structure divides the project into Inception (November 2024), Elaboration (December 2024), Construction (January to March 2025), Transition (March to April 2025), and Maintenance (April 2025). The Gantt chart visualizes dependencies across all phases. Development till date presents system architecture, UML diagrams, flowcharts, wireframes, UI designs, and evidence of completed work including authentication, onboarding, and navigation.

1.6.3. Analysis of Progress

The progress section reviews work against planned milestones. Inception completed on schedule November 30, 2024, delivering business modelling, requirements, risk assessment, and environment setup. Elaboration completed December 31, 2024, producing architecture, UI designs, and working prototypes for WebRTC and offline sync. Construction Iteration 1 is in progress with authentication

ahead of schedule. A progress table compares planned versus actual completion. The action plan addresses upcoming iterations.

1.6.4. Future Work

This section outlines tasks remaining to complete the project. Construction has six more iterations covering content and multimedia (Iteration 2), offline support (Iteration 3), therapist booking and video sessions (Iteration 4), wellness games (Iteration 5), payments and notifications (Iteration 6), and admin systems (Iteration 7). Target completion for Construction is March 15, 2025. Transition runs March 16 to April 15, 2025, handling quality assurance, user acceptance testing, deployment, and documentation. Maintenance runs April 16 to April 30, 2025, covering bug fixes, monitoring, and user support. The conclusion summarizes progress and confirms the project remains on track for completion by April 30, 2025.

2. Background and Literature Review

Understanding who will use Resilio and what they need is essential before building anything. Mental health apps work in sensitive territory. Get the design wrong and people might feel worse. Get it right and the app becomes a bridge between struggling individuals and the support they deserve.

2.1. About the End User / Client

Resilio serves three user groups with a shared goal: making mental health support accessible in Nepal.

- **Primary Users - Individuals Seeking Support**

Nepali adults aged 18 to 45 dealing with stress, anxiety, depression, or emotional difficulties. The Nepal Mental Health Survey found this group carries 12.9 percent lifetime prevalence (Ministry of Health and Population, 2020). Most cannot access traditional services due to distance, cost, or stigma. During Inception, one participant who experienced depression for three years described downloading five apps before giving up. None connected to real therapists. None worked offline.

- **Secondary Users - Mental Health Professionals**

Therapists and counsellors use Resilio to conduct sessions and view client progress. Nepal has only 0.36 psychiatrists per 100,000 people (World Health Organization, 2017). Ms. Sushana Karki described frustration with disconnected apps. Dr. Dibyandra Singh noted 90 percent of patients need support between sessions that current tools fail to provide.

[For client approval letters please see [Appendix F: Client Letter](#)]

- **Tertiary Users - Administrators**

Platform administrators manage content, verify credentials, and monitor engagement.

2.2. Understanding the Project

Resilio integrates professional therapy with self-help tools in one application. Six core areas: authentication, therapist booking with real-time calendars, live WebRTC video sessions, mood tracking following ecological momentary assessment principles (Shiffman et al., 2008), validated questionnaires including PHQ-9 and GAD-7, and wellness content with games and therapist videos.

Technical stack uses Flutter for mobile, Node.js backend, PostgreSQL for server storage, SQLite for offline data. eSewa handles payments. AES-256 encryption protects user data.

[For detailed feature breakdown please see [Appendix D: Detailed Solution Features](#)].

2.3. Review of Similar Projects

Three existing mental health applications were analysed to identify strengths, limitations, and gaps.

2.3.1. BetterHelp - Online Therapy Platform

Founded 2013 in California, the world's largest online therapy platform with 30,000 therapists serving 3 million users (BetterHelp, 2023). Provides therapist matching, video sessions, and messaging. Research by Imel et al. (2020) found text therapy produced significant symptom reductions. However, subscriptions cost \$60-90 weekly, require credit cards, operate only in English, and have no offline functionality.

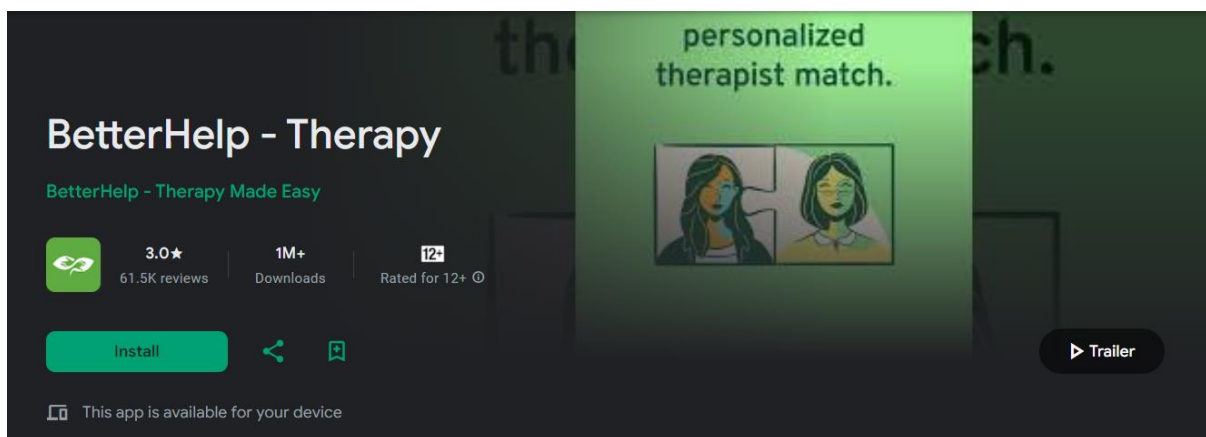


Figure 7: Better Help App

2.3.2. Headspace - Meditation and Mindfulness App

Launched 2010 in UK with 70 million users (Headspace, 2023). Offers structured meditation courses and sleep content. Economides et al. (2018) found regular use improved wellbeing. But no professional therapy connection exists. Costs \$70 yearly. Content primarily English. No mood tracking integration.

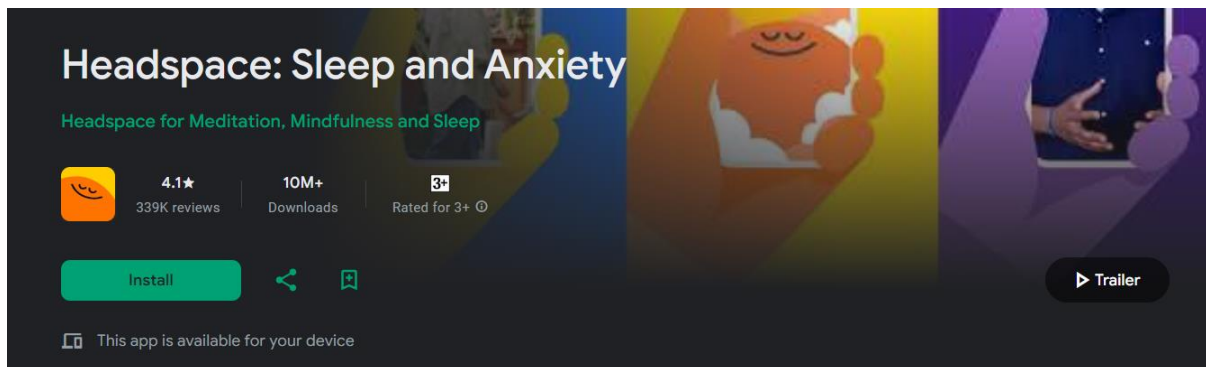


Figure 8: Headspace - Meditation and Mindfulness App

2.3.3. Wysa - AI Mental Health Support App

Indian AI-powered app launched 2017 with 5 million downloads (Wysa, 2023). Chatbot uses CBT and DBT techniques. Inkster et al. (2018) found significant depression reductions. South Asian origin provides better cultural fit. However, AI cannot replace clinical judgment. Human therapists cost extra and cannot access mood data.

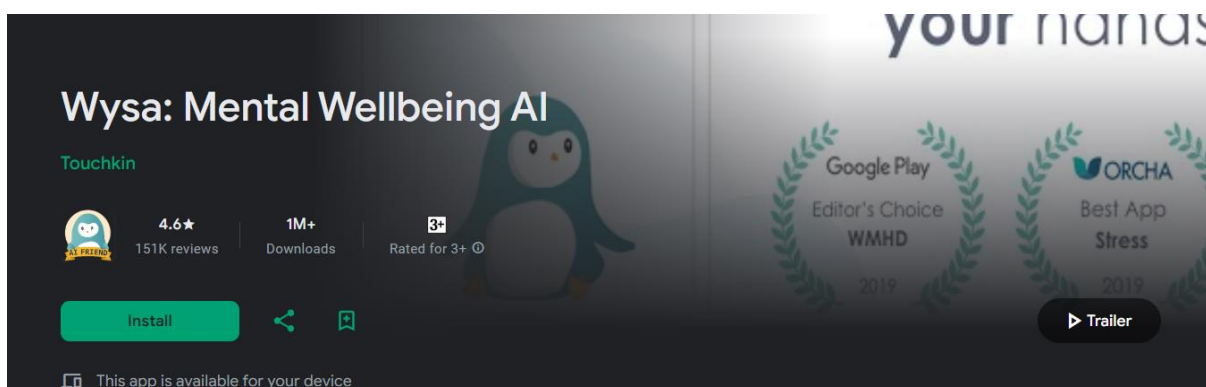


Figure 9: Wysa - AI Mental Health Support App

2.4. Comparison Table of Similar Projects

Feature	BetterHelp	Headspace	Wysa	Resilio
Primary Focus	Online therapy	Meditation	AI support	Integrated wellness
Professional Therapy	Yes	No	Paid add-on	Yes - integrated
Self-Help Tools	Limited	Extensive meditation	CBT exercises	Mood tracking, games, multimedia
AI Chatbot	No	No	Yes - primary	No
Mood Tracking	No	No	Basic	Comprehensive EMA-based
Gamification	None	Basic streaks	Basic streaks	Breathing games, trivia, RPG
Offline Functionality	None	Partial	Partial	Comprehensive
Live Video Sessions	Yes	No	Paid add-on	Yes - WebRTC
Payment Methods	Credit card	Credit card	Credit card	ESewa mobile wallet
Annual Cost (USD)	960-1440	70	Free + 60-100	20-30 estimated
Language Support	English	English mainly	English mainly	Nepali and English
Cultural Adaptation	Western	Western	Some South Asian	Nepal-specific
Target Geography	Global (US/UK focus)	Global	India/global	Nepal
Therapist Data Access	Core feature	N/A	Minimal	Full integration
Founded	2013, USA	2010, UK	2017, India	2025, Nepal

Platform	Web, iOS, Android	iOS, Android	iOS, Android, Web	Flutter (iOS, Android)
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2.5. Comparison Analysis of Similar Projects

The analysis reveals patterns that directly informed Resilio's design.

Integration vs Fragmentation

All three apps operate in isolation. BetterHelp provides therapy without self-help tools. Headspace offers meditation without professional connection. Wysa treats human therapists as separate add-ons. (Bakker et al., 2016) found users abandon fragmented apps within weeks. Resilio combines everything in one platform.

Cultural Adaptation Gaps

Western apps ignore Nepali context. (Kohrt and Harper, 2008) documented how Nepalis understand mental distress through spiritual and social dimensions. BetterHelp costs exceed annual healthcare spending for many families. Resilio addresses this through Nepali language, local therapists, eSewa integration, and \$20-30 annual pricing with free tier.

Connectivity Barriers

BetterHelp has no offline capability. Only 15 percent of apps use gamification despite 40-60 percent retention improvement (Fleming et al., 2019). Resilio's offline architecture and gamified features address both gaps.

Economic Barriers

BetterHelp costs \$2,880 to \$4,320 annually. Headspace costs \$70. All require credit cards. (Luitel et al., 2015) identified cost as the primary barrier preventing most Nepalis from seeking treatment. Resilio addresses this through eSewa integration, local pricing around \$20-30 annually, and a free tier for self-help tools.

[For detailed feature breakdown please see Appendix C: Detailed App Comparison].

3. Development Methodology

Selecting the right development approach matters as much as selecting the right features. A poorly matched methodology leads to missed deadlines, unclear requirements, or a product that does not meet user needs. For Resilio, three software development lifecycle models were evaluated based on project requirements, academic constraints, and the sensitive nature of mental health applications.

3.1. Considered Methodology

Selecting the right development approach matters as much as selecting the right features. A poorly matched methodology leads to missed deadlines, unclear requirements, or a product that does not meet user needs. For Resilio, three software development lifecycle models were evaluated based on project requirements, academic constraints, and the sensitive nature of mental health applications.



Figure 10: Software Development Lifecycle

3.1.1. Agile Model

Agile emphasizes iterative development, frequent stakeholder feedback, and rapid adaptation to change (Beck et al., 2001). Teams deliver working software in short sprints, typically two to four weeks. This allows flexible requirement updates and continuous delivery.



Figure 11: Agile Model

However, Agile struggles with scope creep when requirements keep expanding. Documentation often receives less attention than working code. For academic projects with fixed deadlines and documentation requirements, aligning sprint cycles with submission milestones becomes challenging. The flexibility that makes Agile powerful in commercial settings creates risks when deliverables must meet specific academic criteria.

3.1.2. Rational Unified Process (RUP)

RUP is an iterative, use-case-driven model divided into four phases: Inception, Elaboration, Construction, and Transition (Kruchten, 2003). It balances structured documentation with iterative development. Risk management is built into early phases, ensuring critical technical challenges are addressed before major construction begins. Stakeholders provide input at defined points throughout the process.

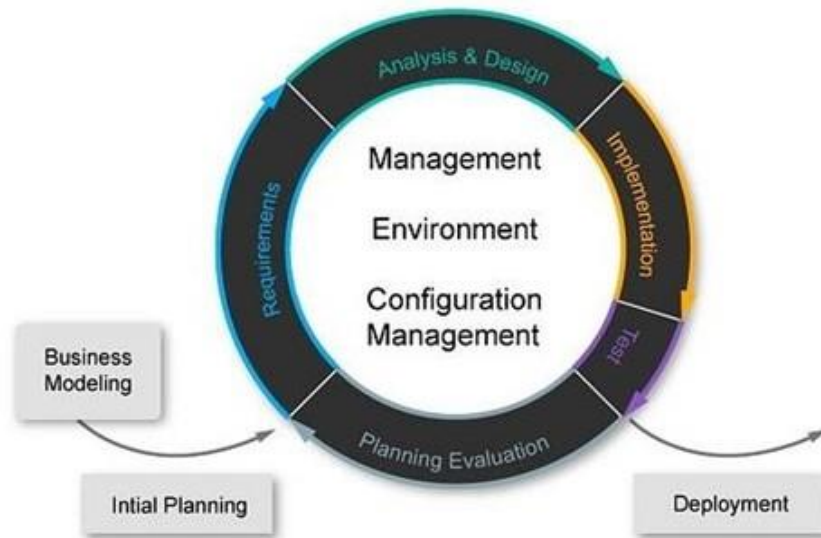


Figure 12: RUP Model

RUP ensures critical system components are validated early while allowing requirement refinement as understanding deepens. Each phase produces specific deliverables that map naturally to academic milestones.

3.1.3. Prototype Model

This approach focuses on building early prototypes for user evaluation (Sommerville, 2016). Users interact with rough versions of the system, providing feedback that shapes subsequent development. It supports rapid visualization of features and early identification of usability issues.

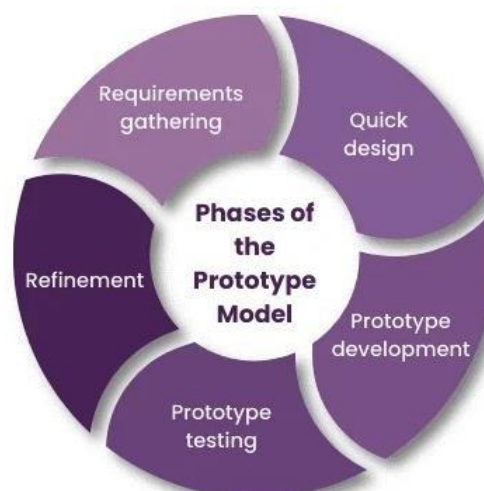


Figure 13: Prototype Model

However, extensive prototyping may increase development effort without clear boundaries. Without strict phase management, projects risk incomplete documentation or misaligned feature scope, which becomes problematic for academic deliverables with defined requirements.

[For detailed model comparison matrix please see [Appendix E: Methodology Details](#)]

3.2. Selected Methodology

RUP was selected for Resilio's development based on several factors aligned with project needs.

- **Iterative Development with Structure**

Mental health applications require careful refinement. A mood tracking feature that frustrates users could discourage engagement entirely. RUP allows phased development with iterative cycles within each phase, supporting continuous refinement of sensitive features like mood tracking and therapist sessions while maintaining clear milestones and documentation.

- **Requirement Clarity**

Resilio serves diverse stakeholders: end users seeking support, therapists providing services, and administrators managing content. RUP's Elaboration phase specifically addresses requirement refinement through stakeholder review of prototypes and use cases. This ensures features like questionnaires, gamified exercises, and motivational content meet actual user needs rather than assumed ones.

- **Risk Management**

Technical challenges including WebRTC integration for video sessions, SQLite offline storage, and eSewa payment processing could derail development if discovered late. RUP addresses high-risk components during Elaboration, validating architecture before major construction begins (Kruchten, 2003). Problems surface early when they are cheaper to fix.

- **Stakeholder Engagement**

Each RUP phase includes defined points for user evaluation and expert review. Mental health professionals validated clinical appropriateness of questionnaires and crisis protocols. Potential users tested interface designs. This structured engagement ensures the final product is both technically sound and clinically safe.

- **Alignment with Academic Timeline**

RUP phases map naturally to academic milestones. Inception aligned with proposal submission in November 2024. Elaboration aligned with interim report in December 2024. Construction runs through development phase from January to March 2025. Transition aligns with final submission in April 2025. This alignment ensures measurable progress against deadlines without forcing artificial sprint boundaries.

3.3. Work Breakdown Structure

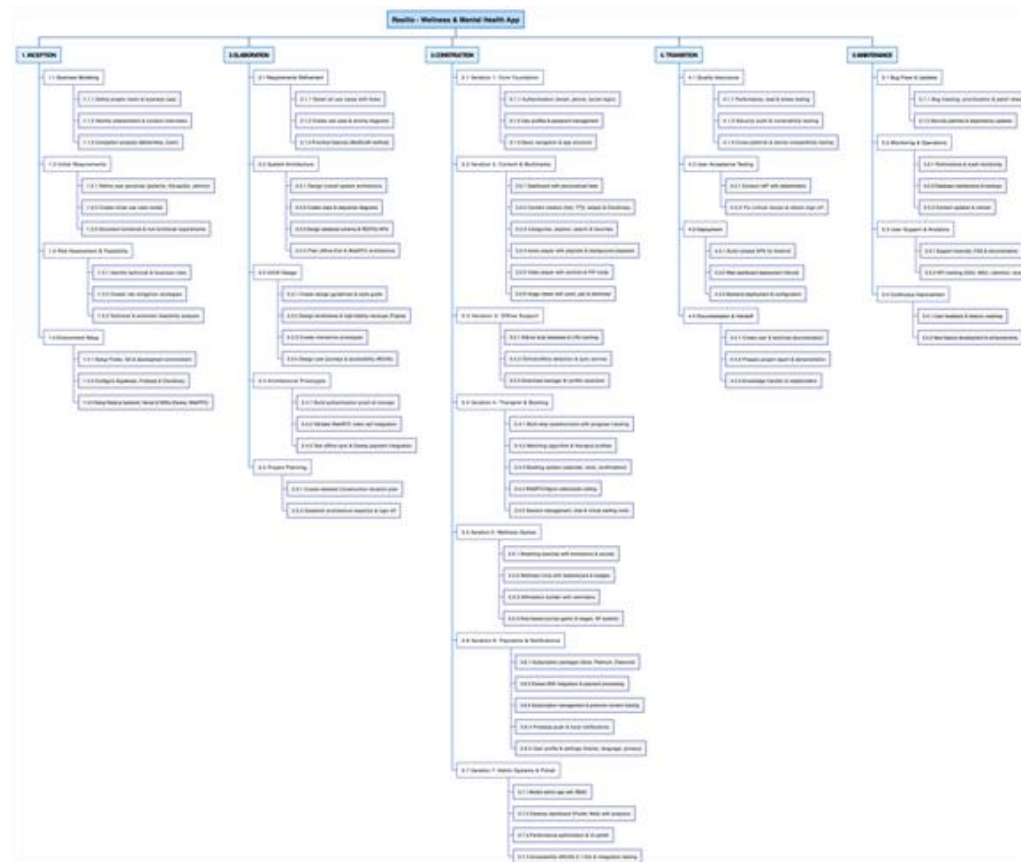


Figure 14: Work Breakdown Structure

[For a clear view of WBS, please [Click Here](#)]

3.4. Milestones



Figure 15: Milestone Timeline Chart

3.5. Project Gantt Chart

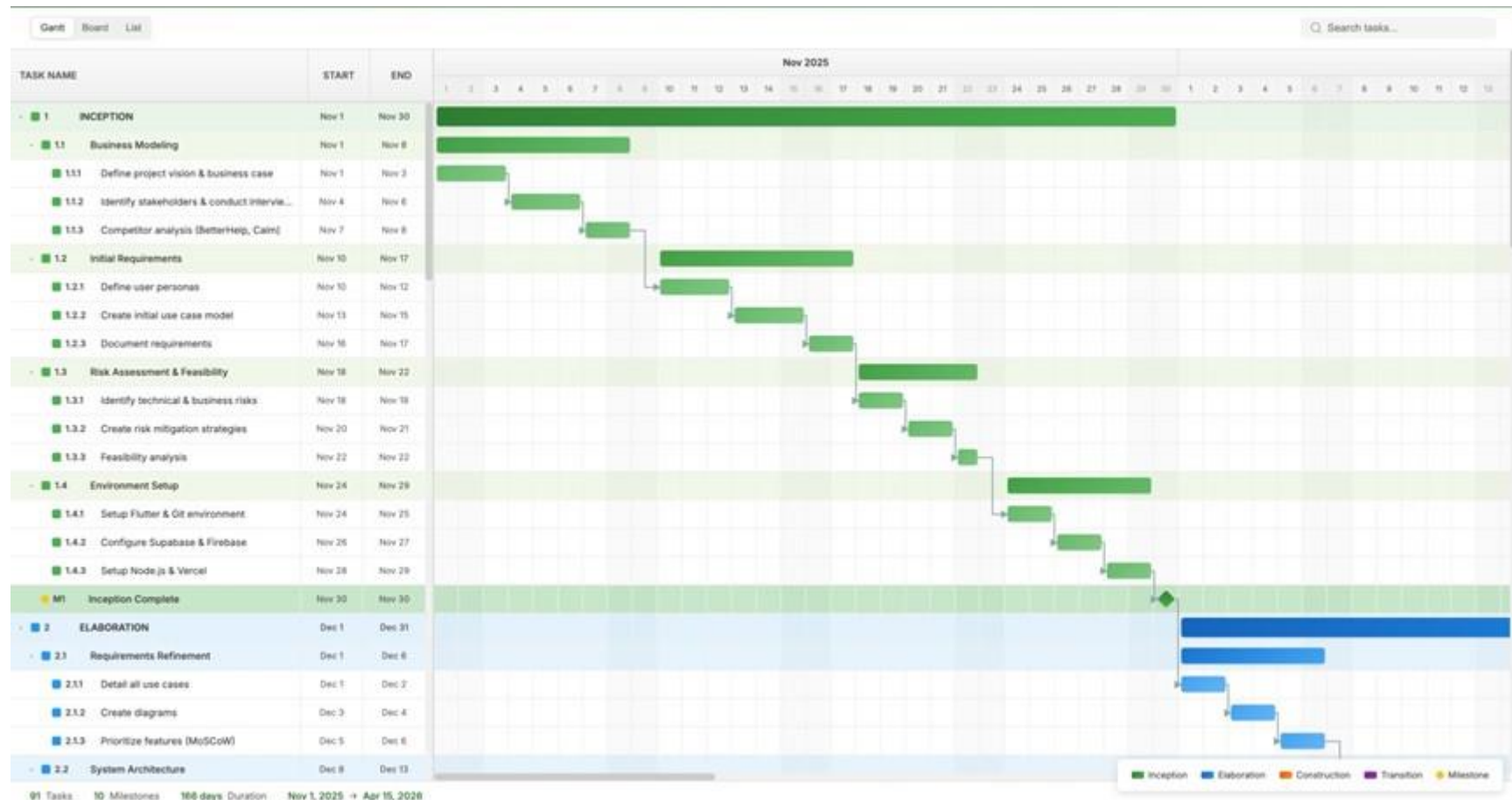


Figure 16: Gantt Chart-1

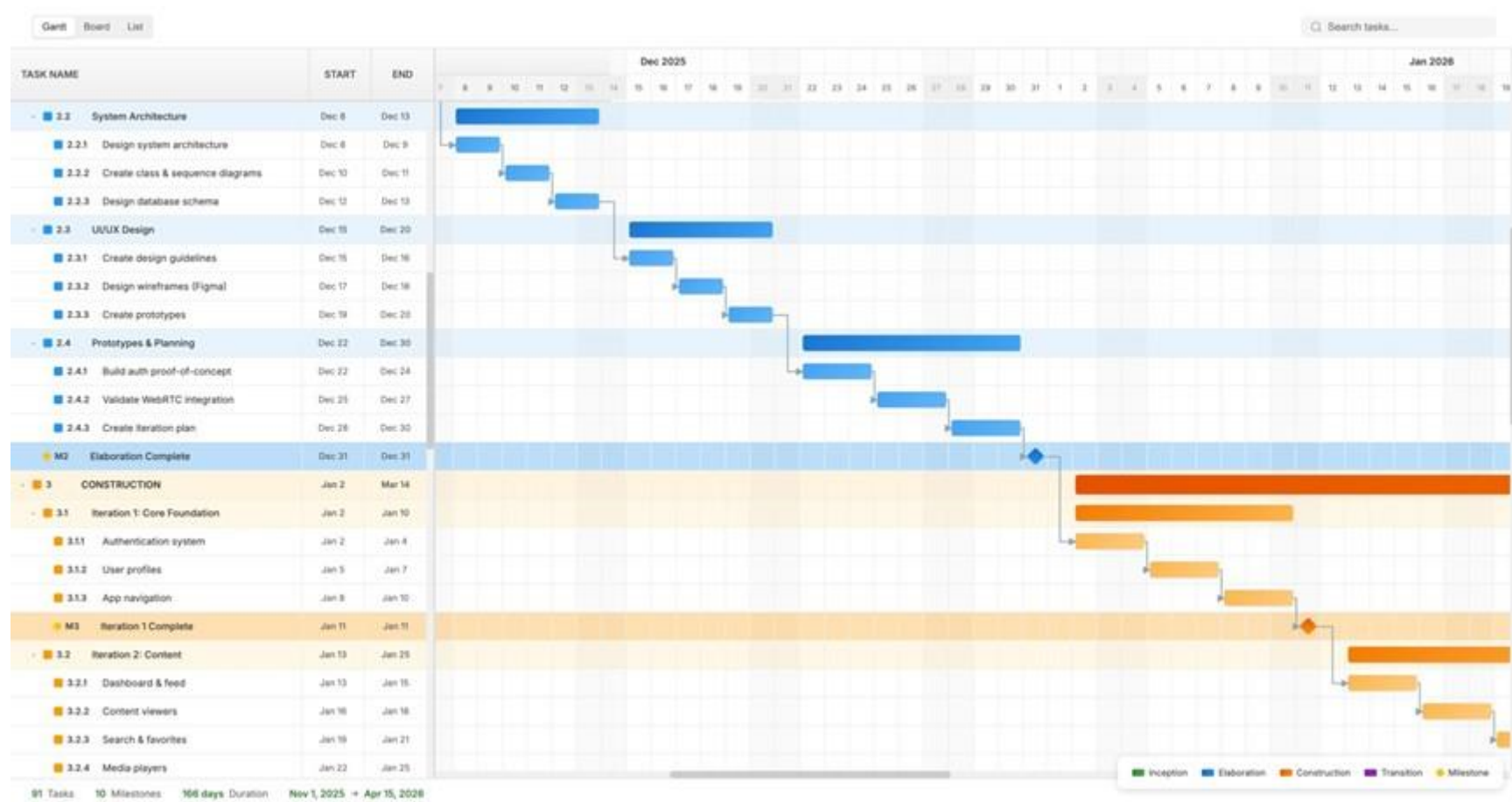


Figure 17: Gantt Chart -2

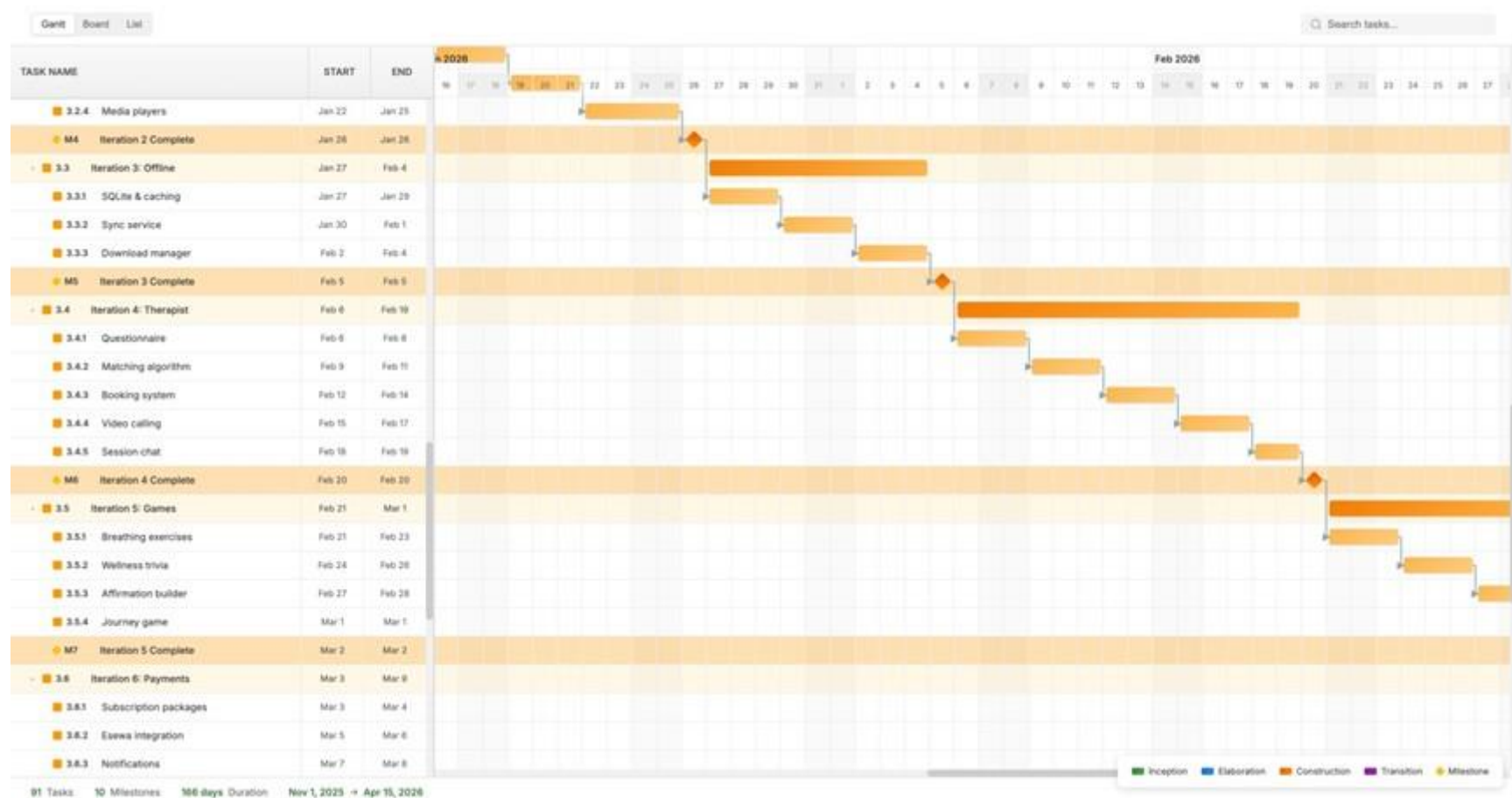


Figure 18: Gantt Chart -3

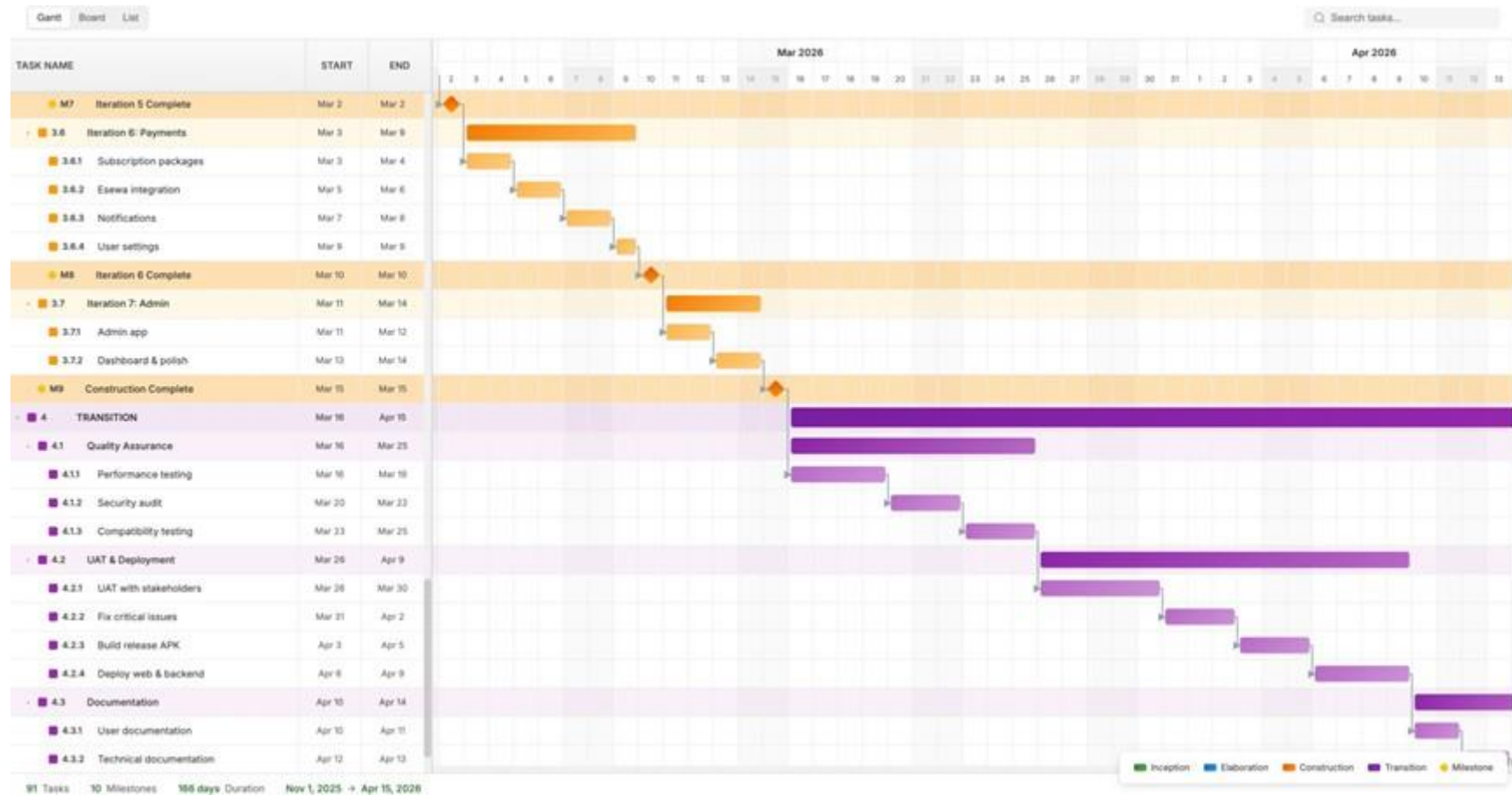


Figure 19: Gantt Chart -4

4. Development till Date

This section reviews development progress against the planned timeline established in the work breakdown structure. The project follows Rational Unified Process methodology with defined milestones for each phase.

4.1. System Architecture Diagram

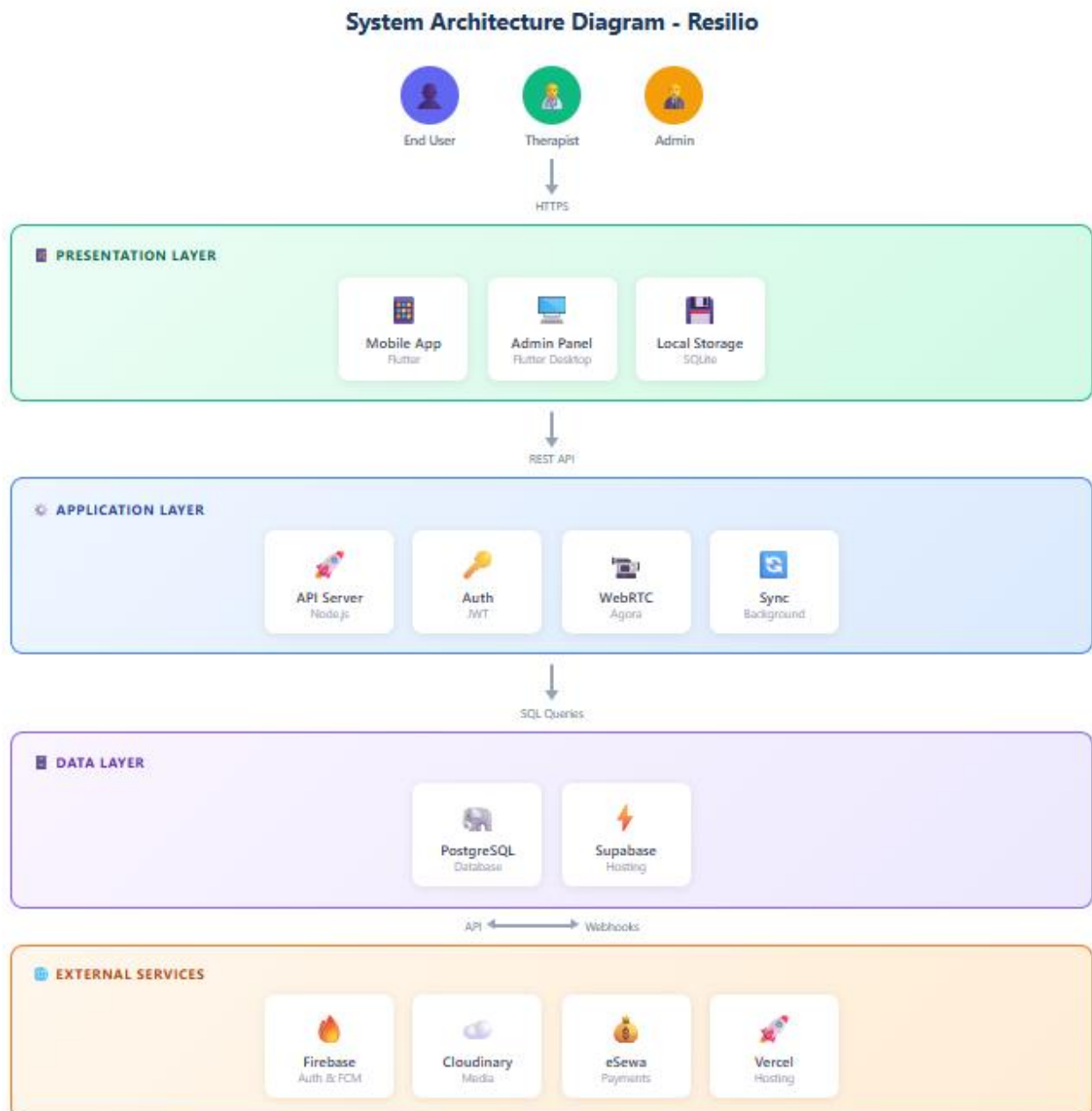


Figure 20: System Architecture Diagram

4.2. Block Diagram

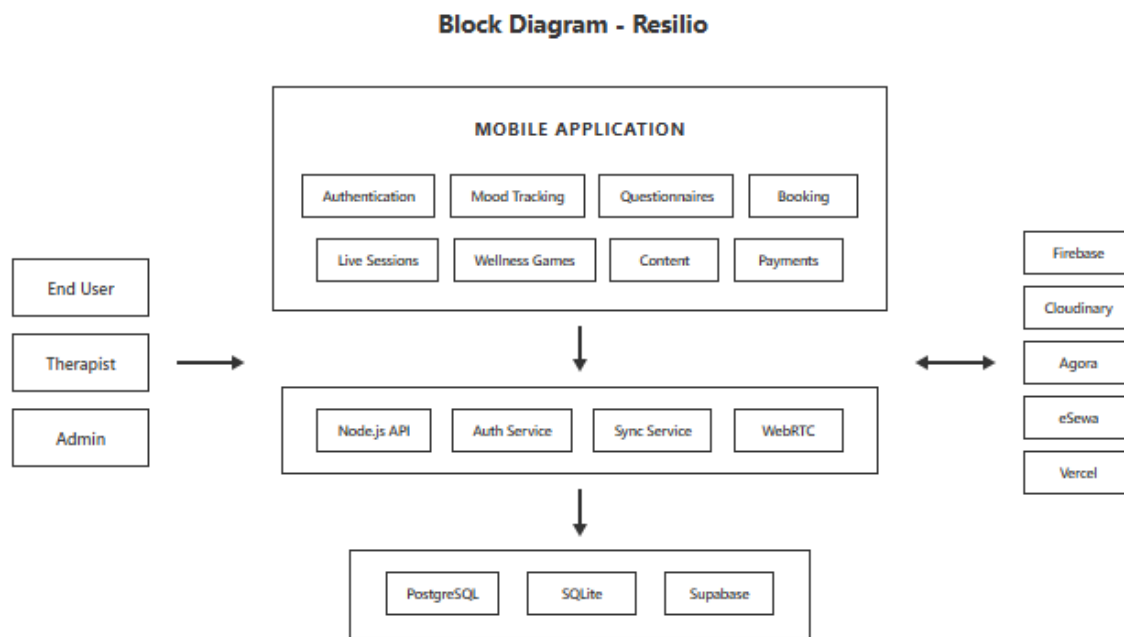
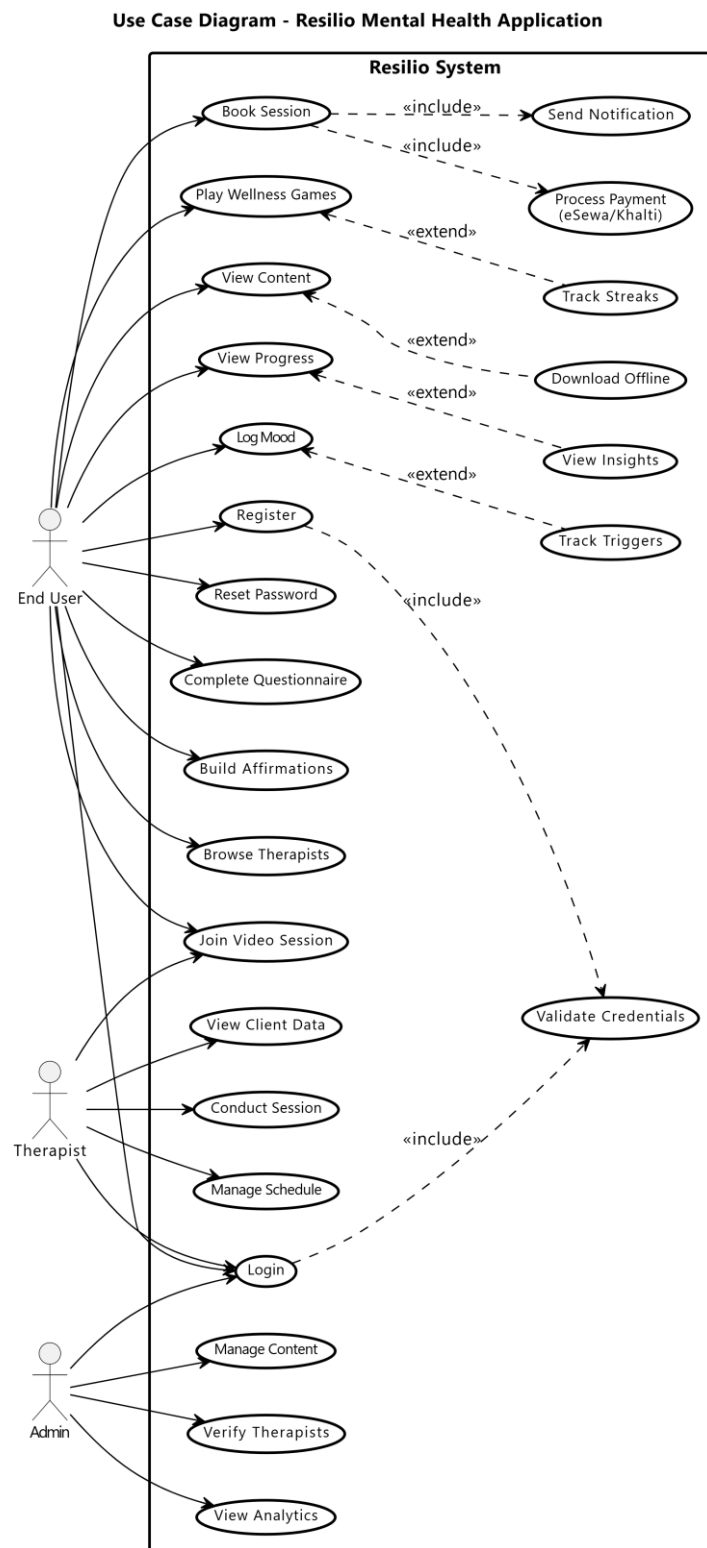


Figure: Block Diagram

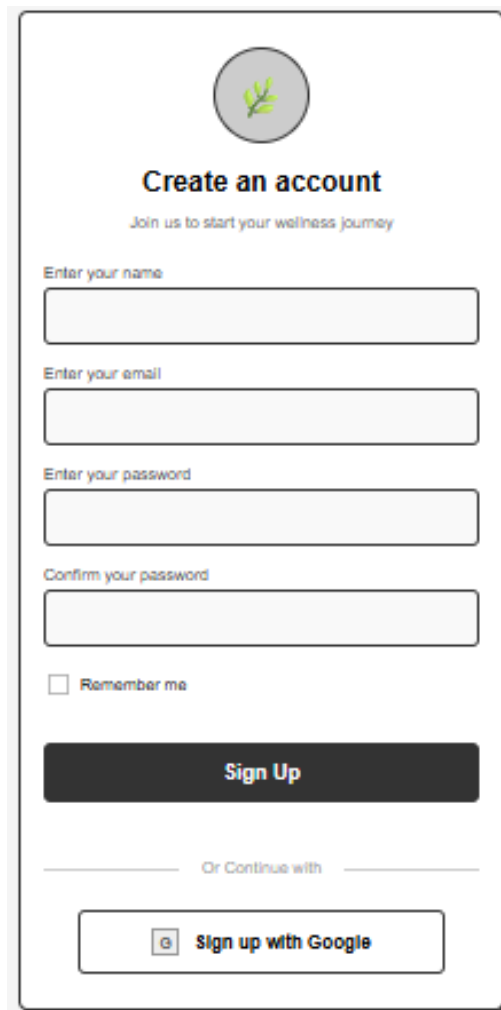
Figure 21: Block Diagram

4.3. Use Case Diagram

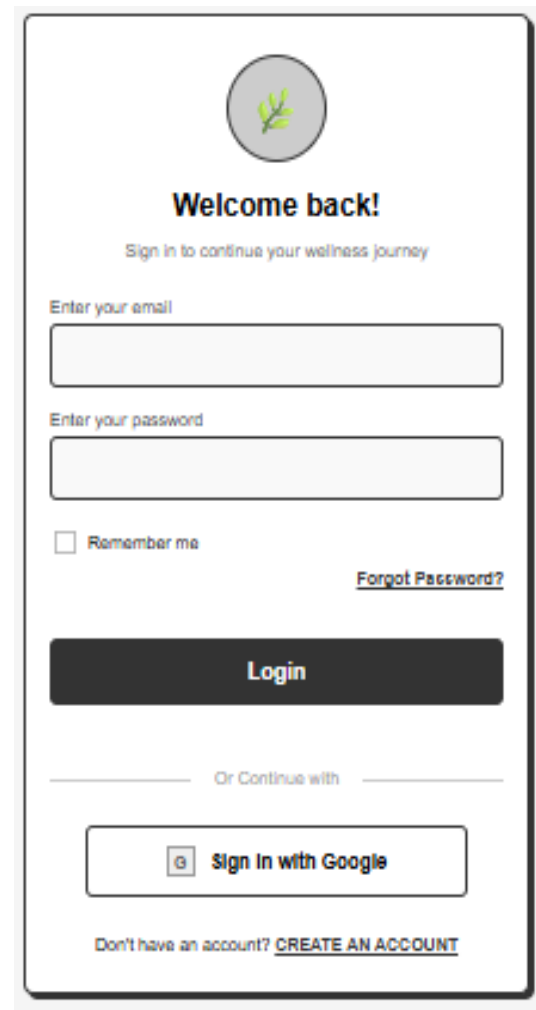


4.4. Wireframe

4.4.2. Login Screen

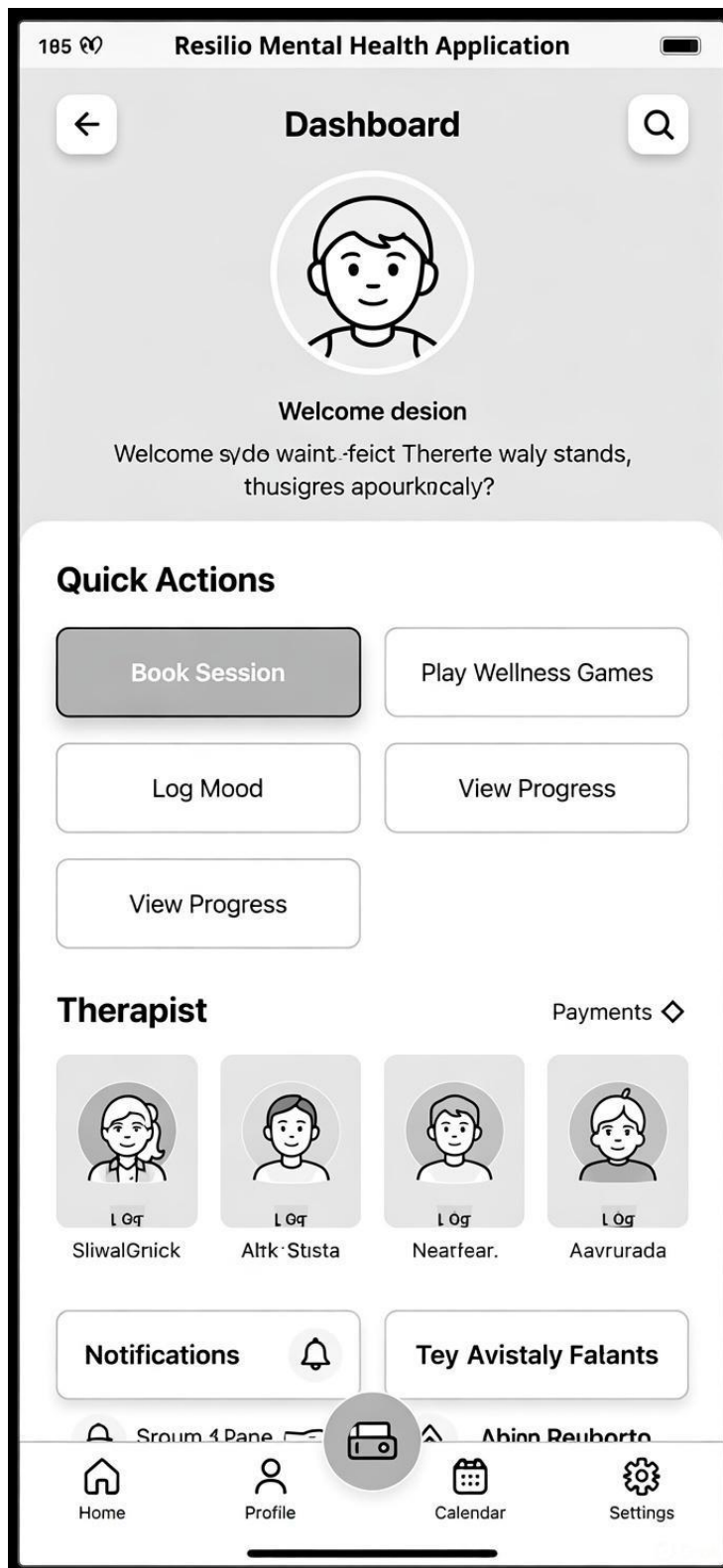


Wireframe for the 'Create an account' screen. It features a circular logo with a green leaf icon at the top. Below the logo is the title 'Create an account' and the subtitle 'Join us to start your wellness journey'. The form includes four input fields: 'Enter your name', 'Enter your email', 'Enter your password', and 'Confirm your password'. There is a 'Remember me' checkbox and a 'Sign Up' button. At the bottom, there is a link 'Or Continue with' followed by a 'Sign up with Google' button.



Wireframe for the 'Welcome back!' login screen. It features a circular logo with a green leaf icon at the top. Below the logo is the title 'Welcome back!' and the subtitle 'Sign in to continue your wellness journey'. The form includes two input fields: 'Enter your email' and 'Enter your password'. There is a 'Remember me' checkbox and a 'Forgot Password?' link. A 'Login' button is positioned below the input fields. At the bottom, there is a link 'Or Continue with' followed by a 'Sign in with Google' button. At the very bottom, there is a link 'Don't have an account? CREATE AN ACCOUNT'.





4.5. UI Design

4.5.1. Onboarding Screen

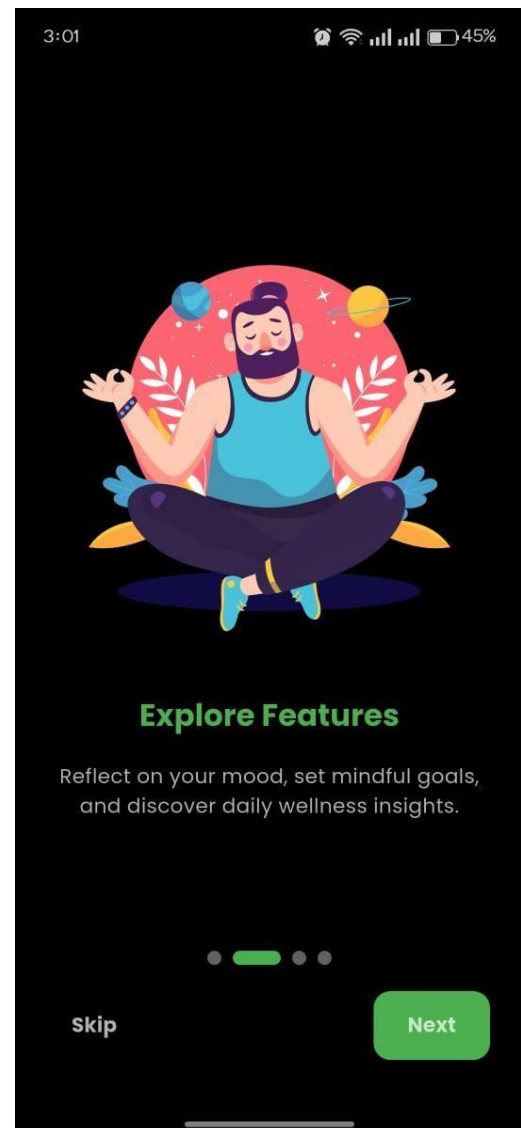
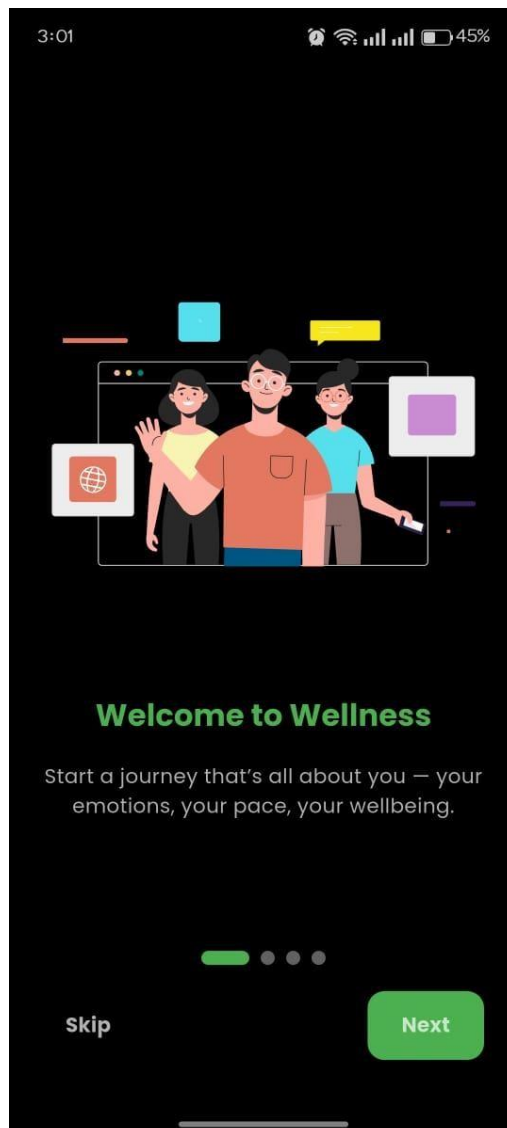


Figure 22: Onboarding Screen -1

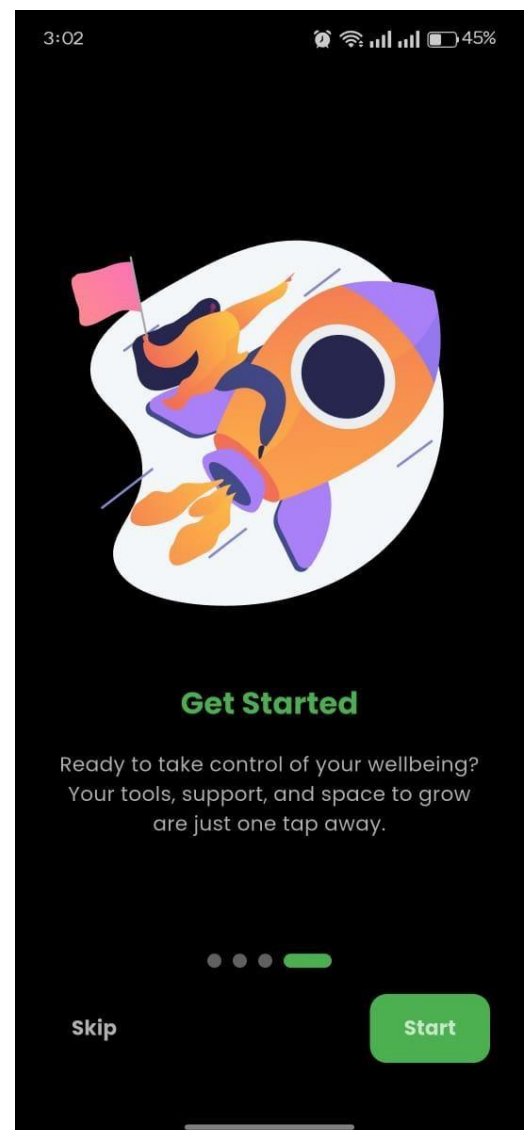
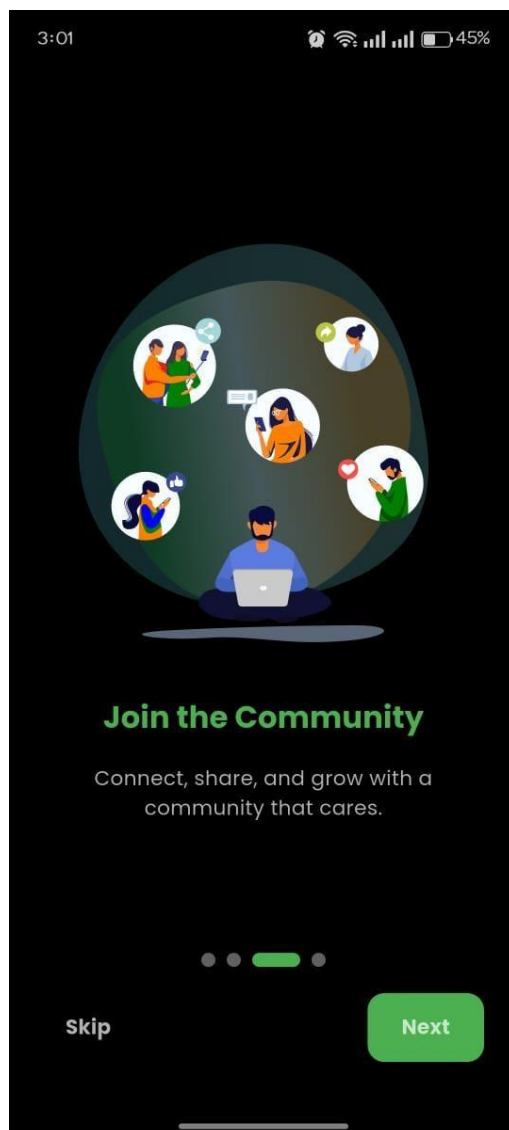


Figure 23: Onboarding Screen -2

4.5.2. Login Screen

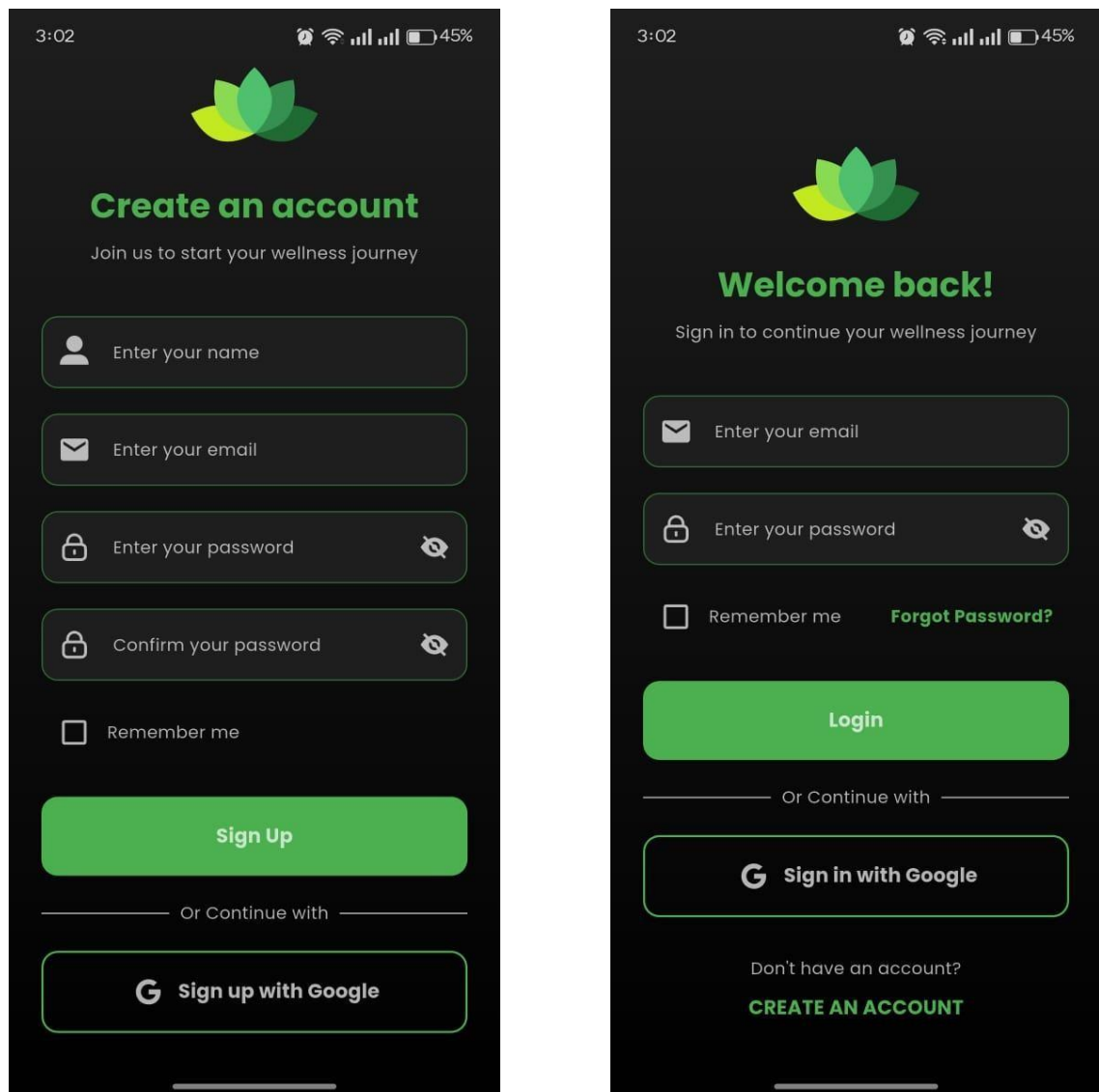


Figure 24: Login Screen

4.5.3. User Dashboard Screen (Unsubscribed User)

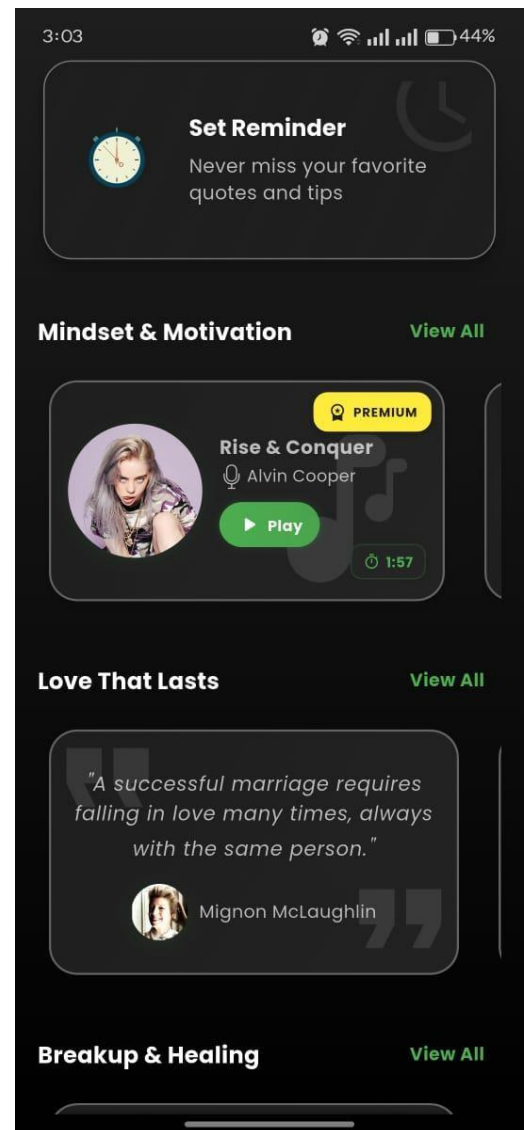
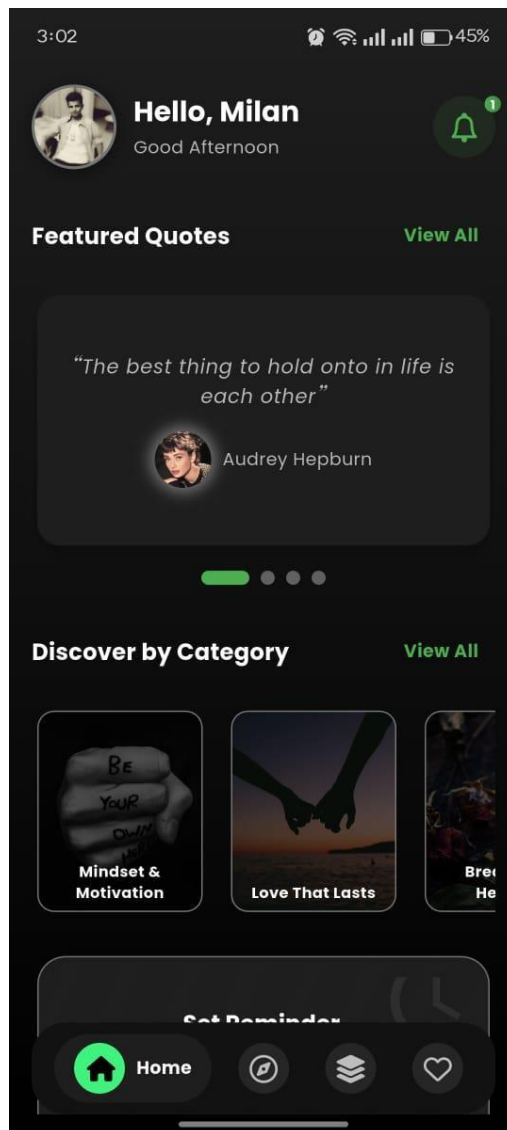
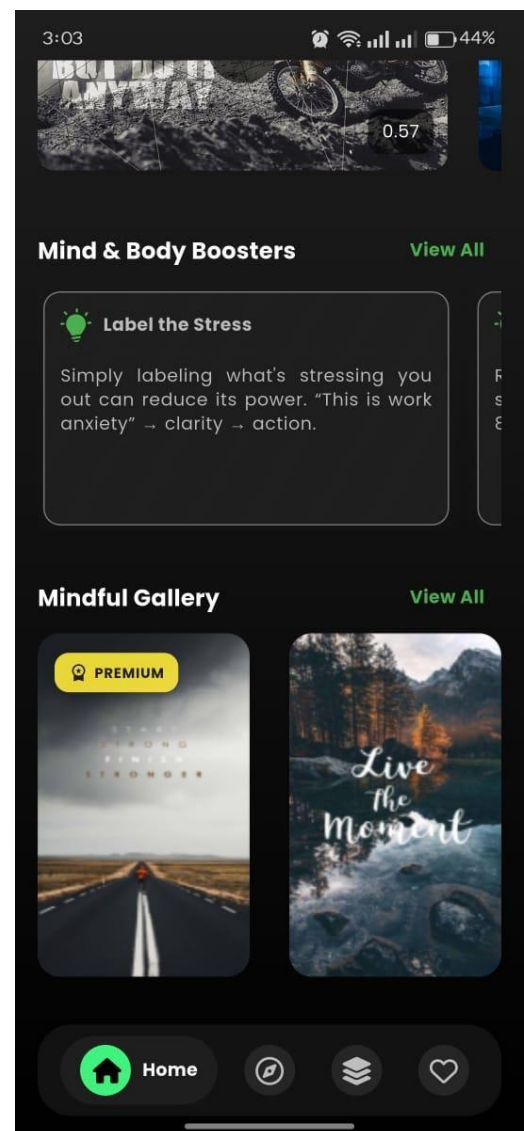
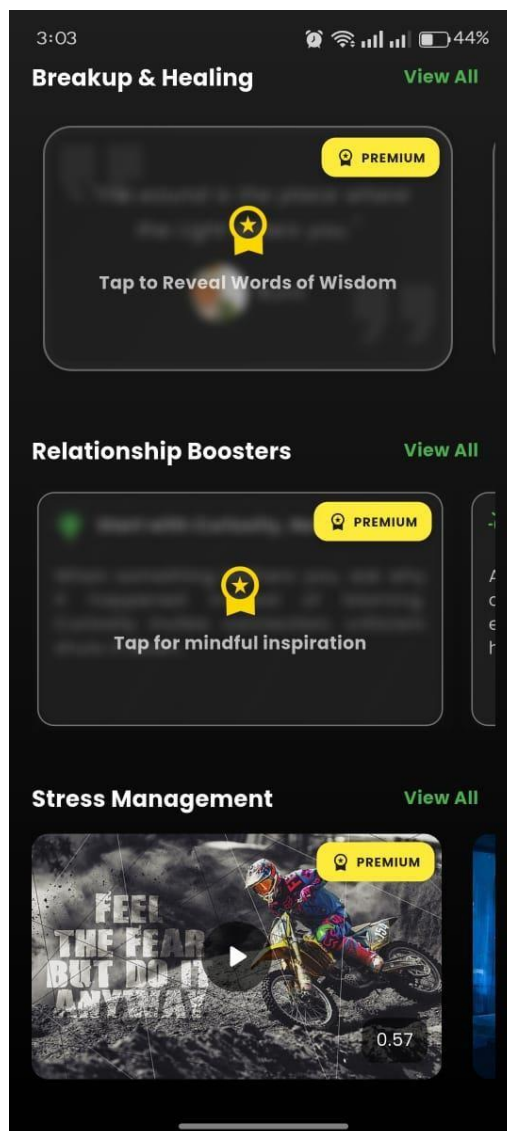


Figure 25: User Dashboard Unsubscribed User -1



4.5.4. Explore Screen

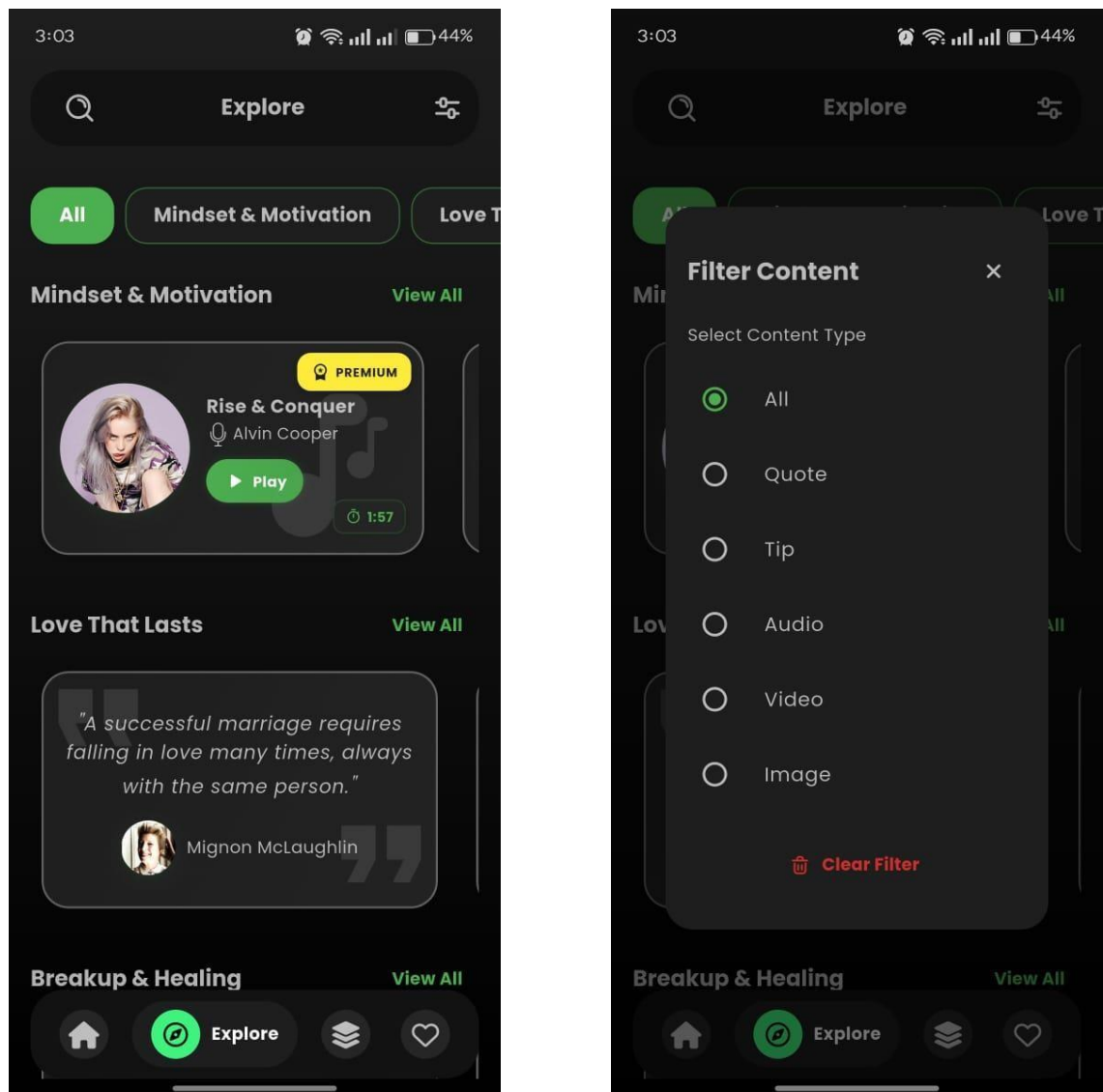
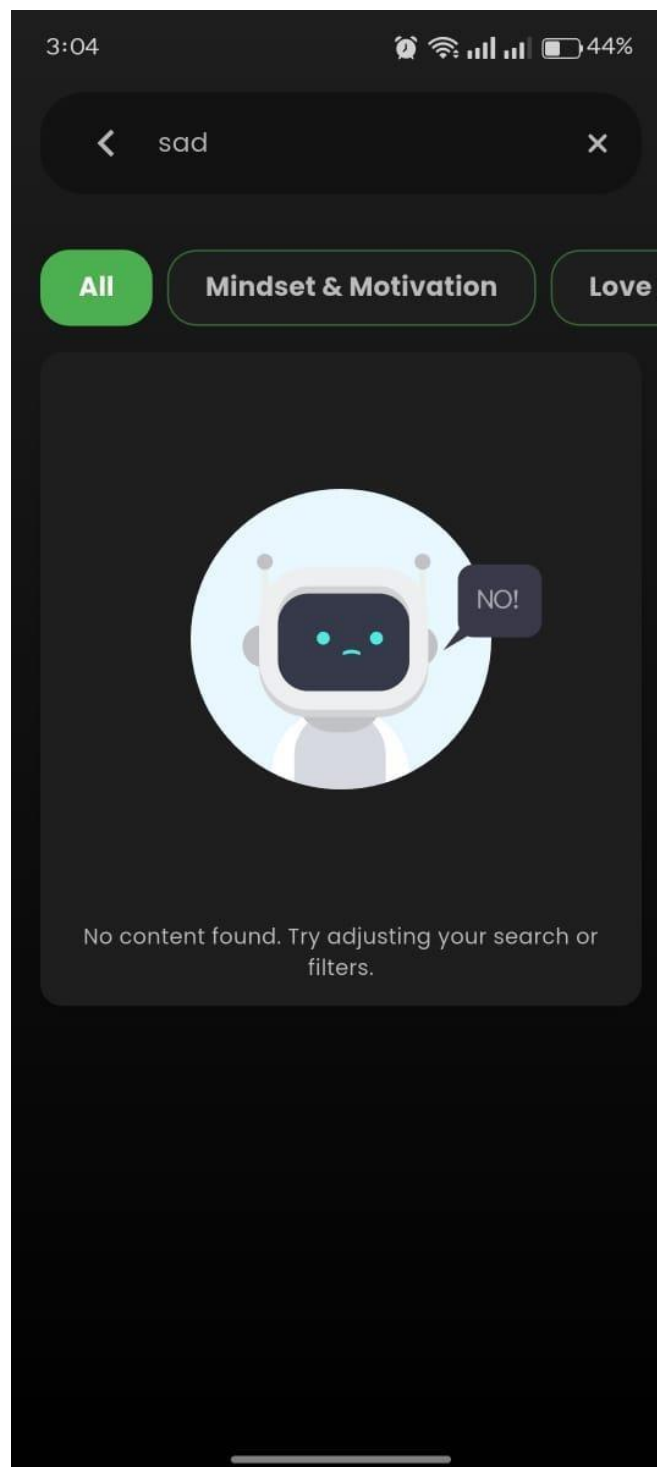
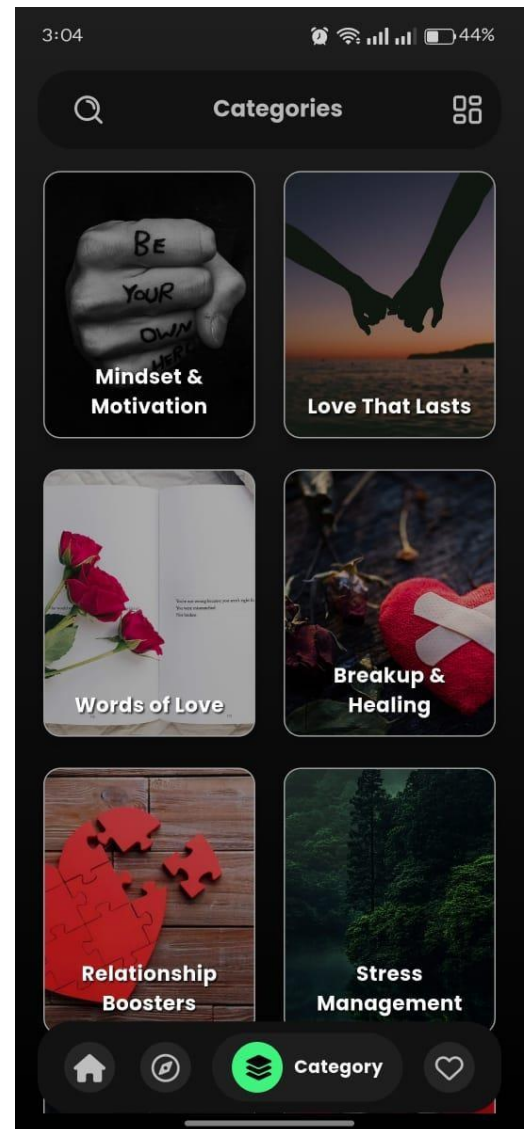
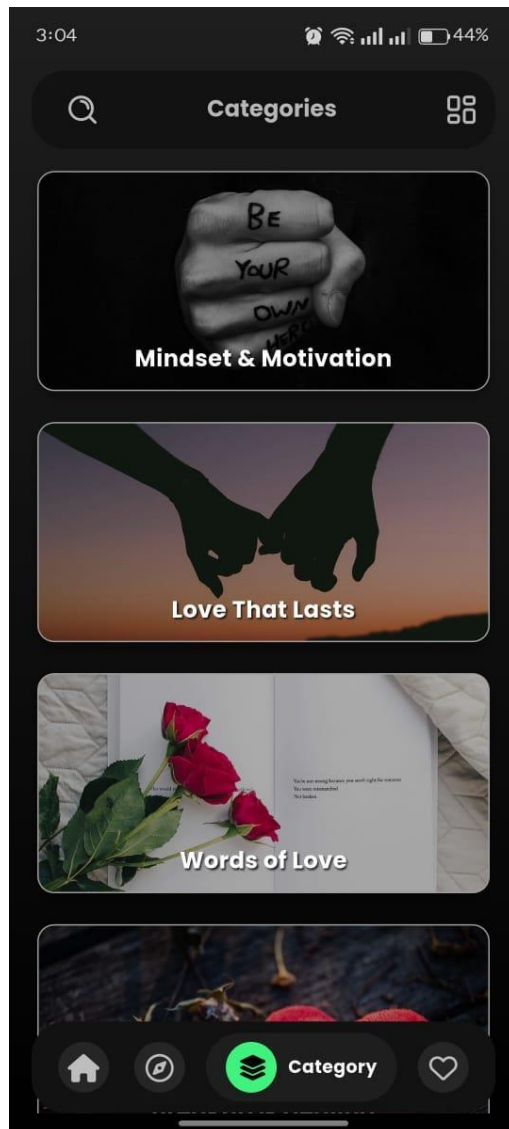


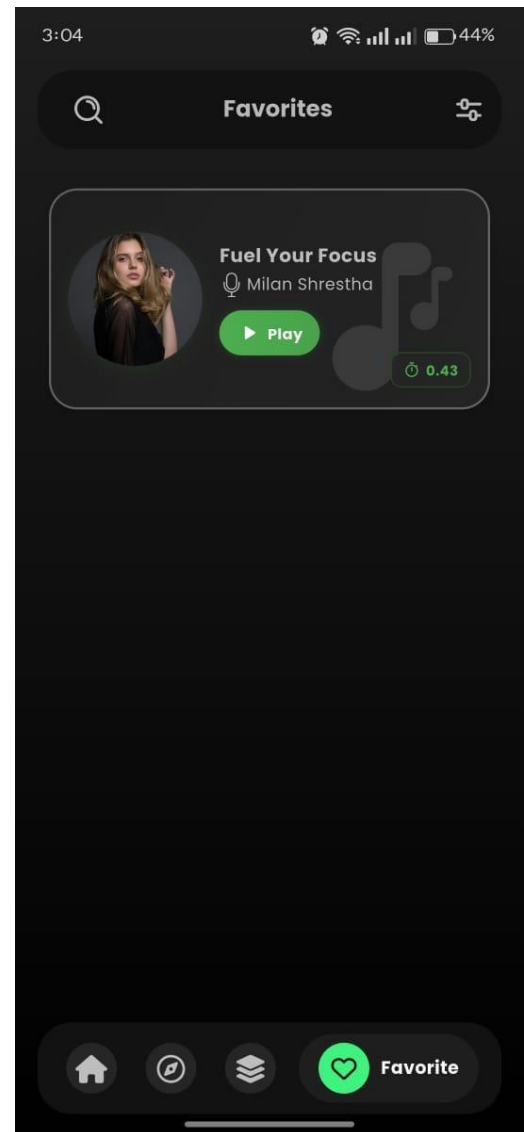
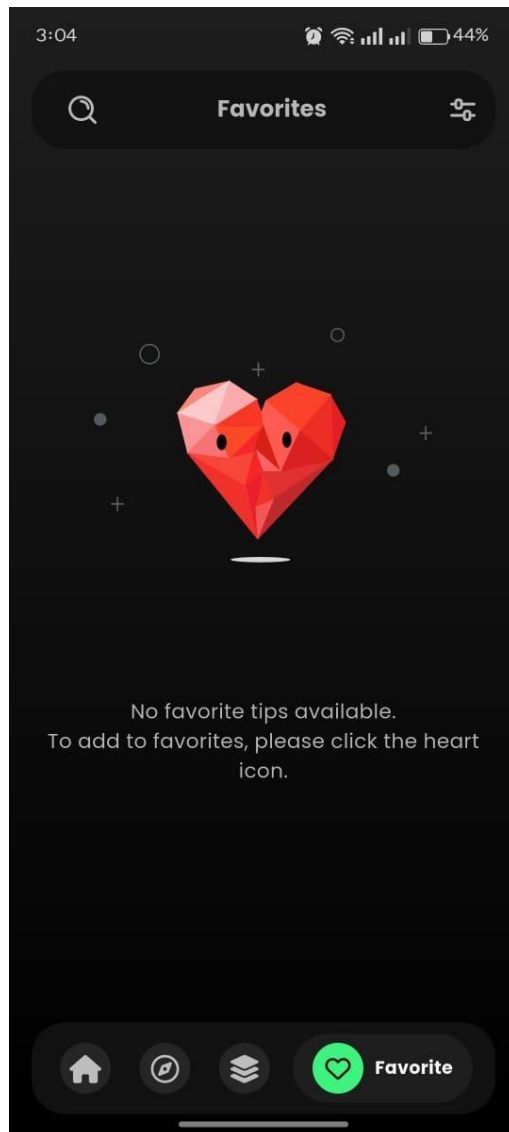
Figure 26: Explore Screen



4.5.5. Category Screen



4.5.6. Favourite Screen



4.5.6. Reminder Screen

3:05 44%

< Create Reminder ↺

Content Type

Select Content Type ▾

Category

All Categories ▾

Frequency

Select Frequency ▾

Time

Select Time ⌚

Save Reminder

3:05 44%

< Reminder History ↺ +

Audio Reminder ✎

Category: Mindset & Motivation

Time: 23:35

Created: 2025-08-19 Active

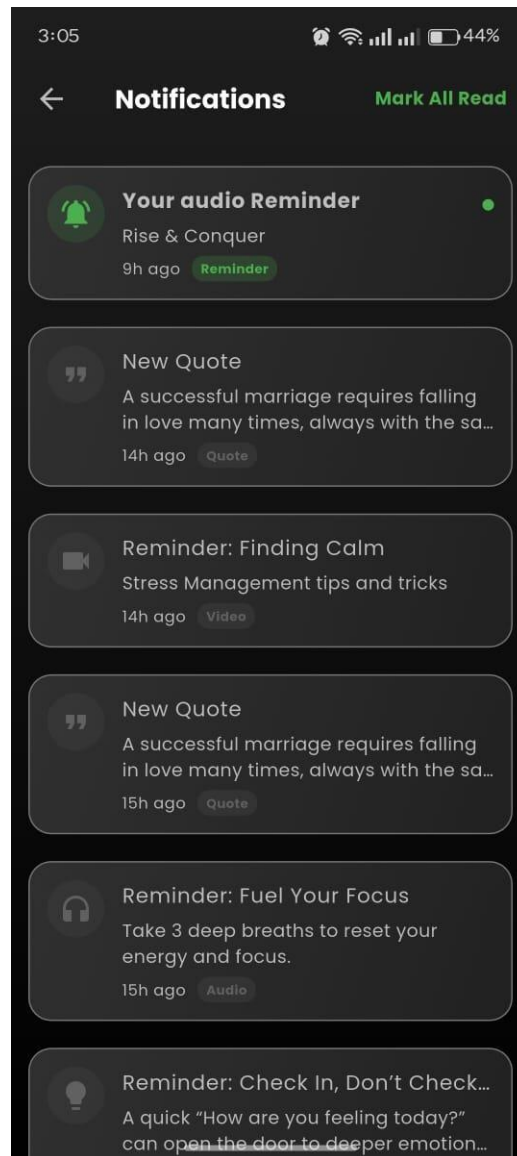
Video Reminder ✎

Category: Stress Management

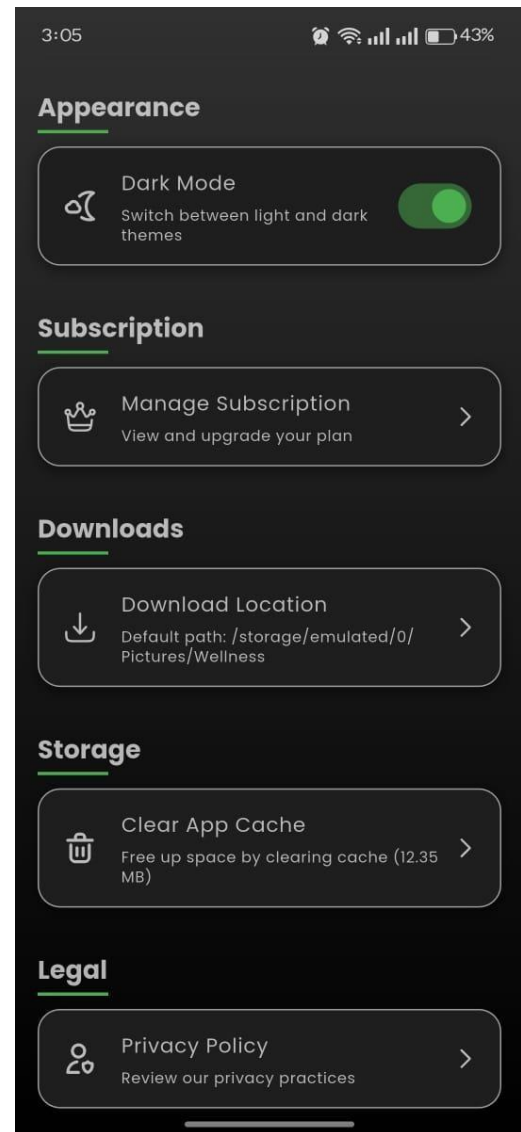
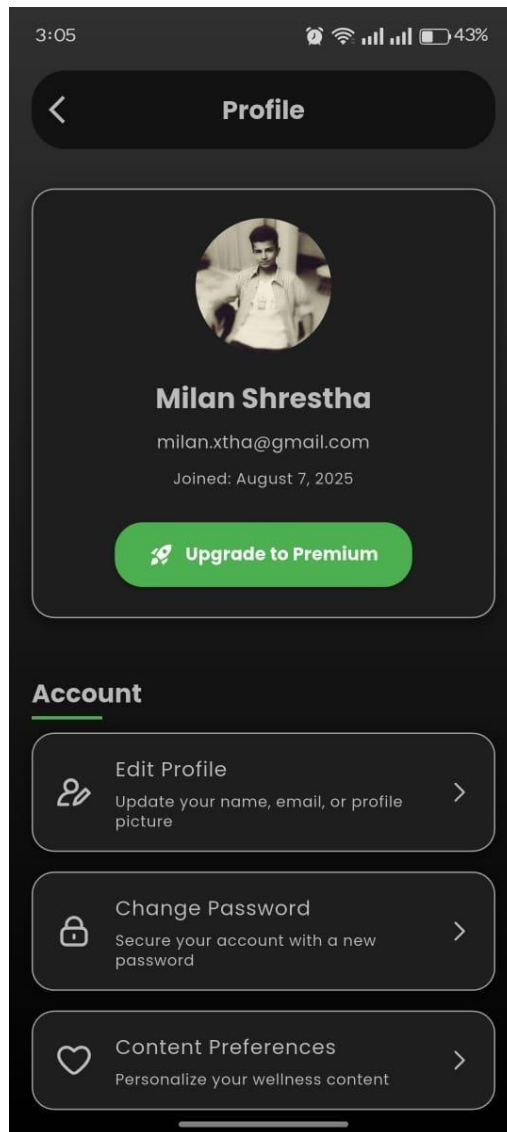
Time: 00:06

Created: 2025-08-20 Active

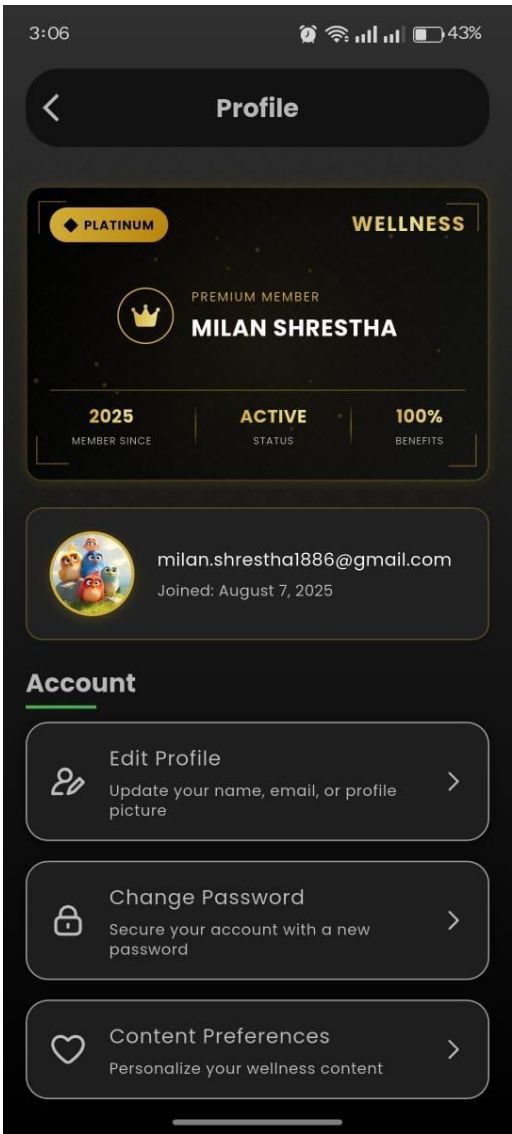
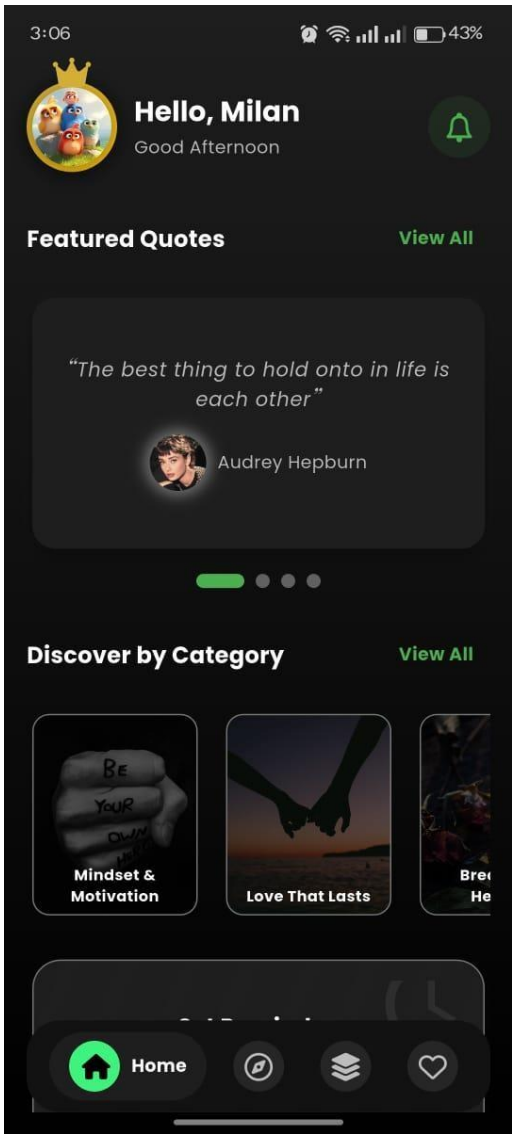
4.5.7. Notification Screen



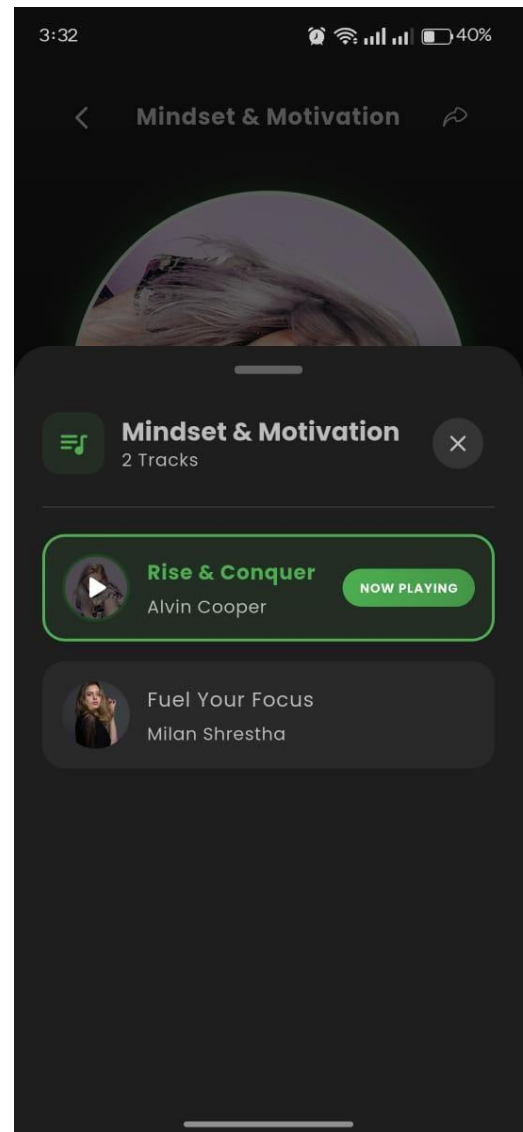
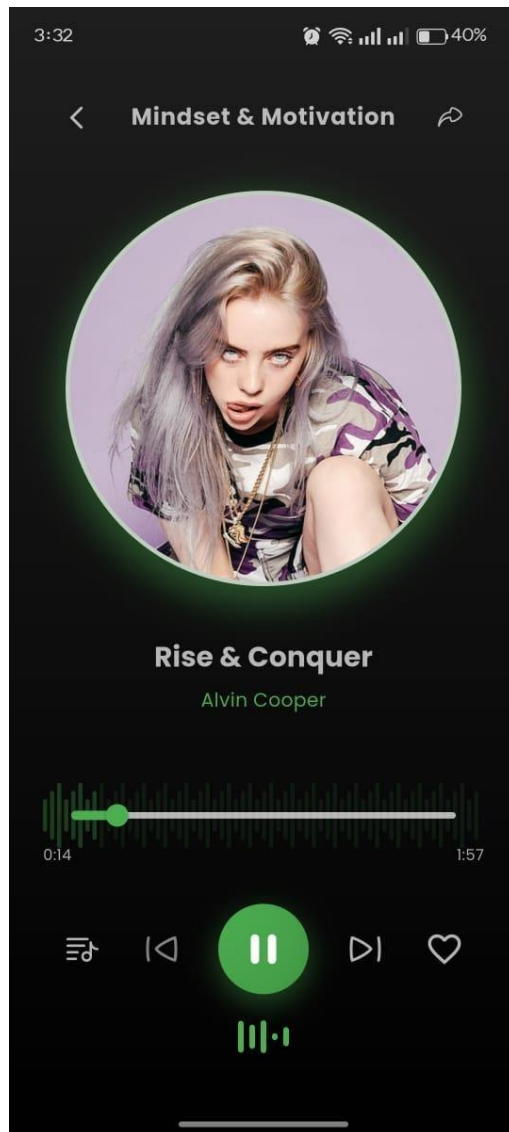
4.5.8. Profile Screen



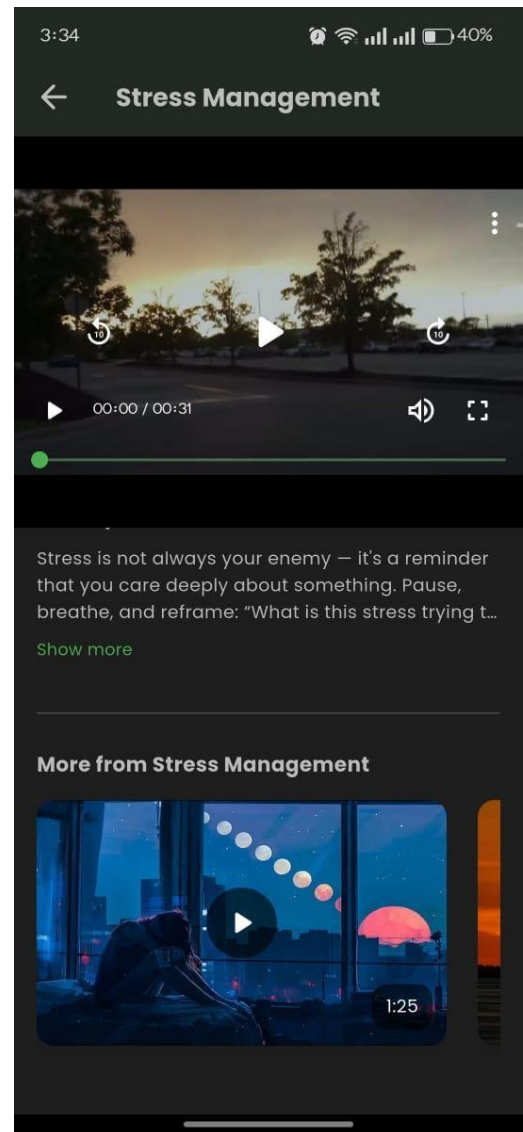
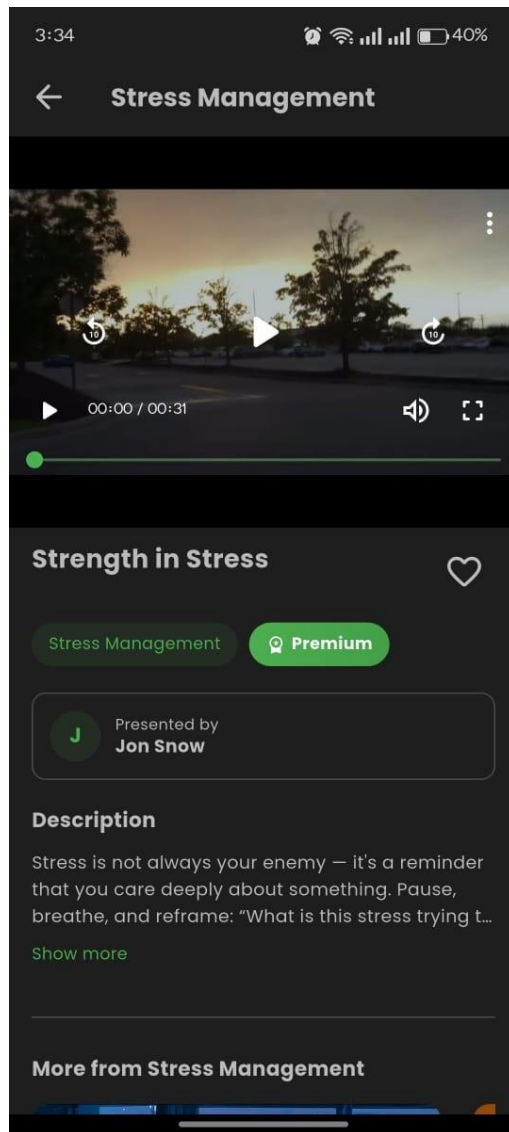
4.5.9. Premium User Screen



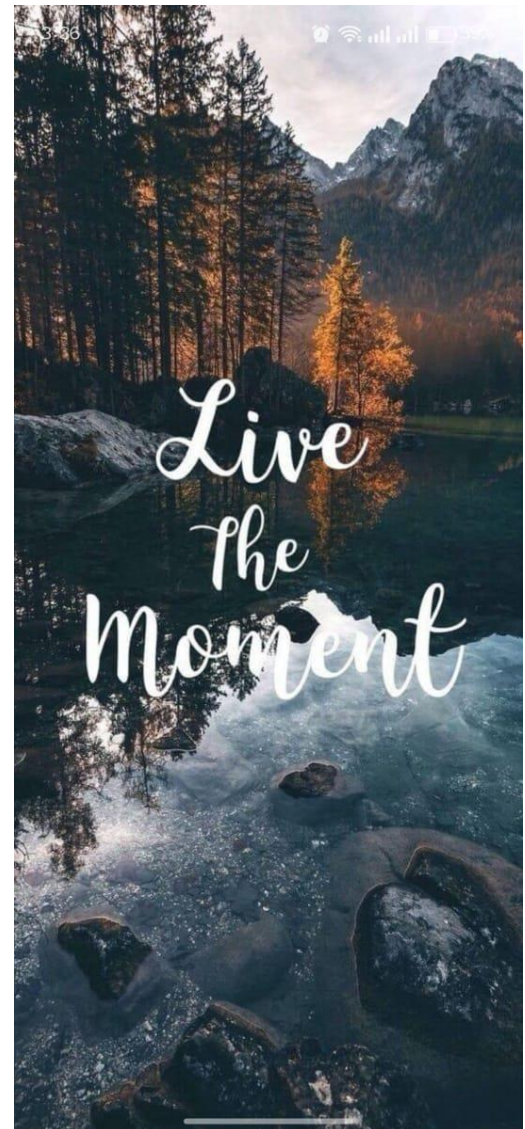
4.5.10. Media Player Screen



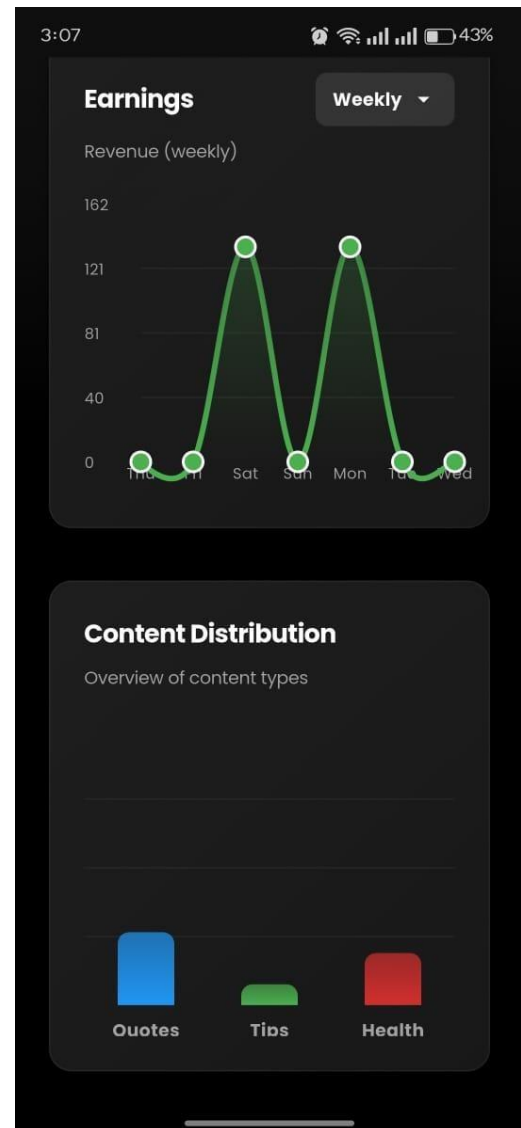
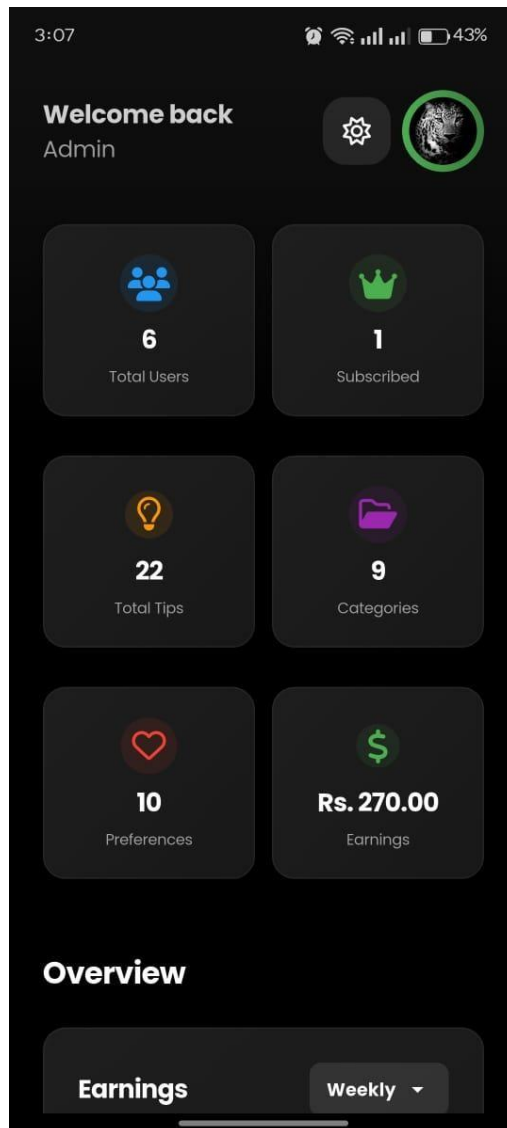
4.5.11. Video Player Screen

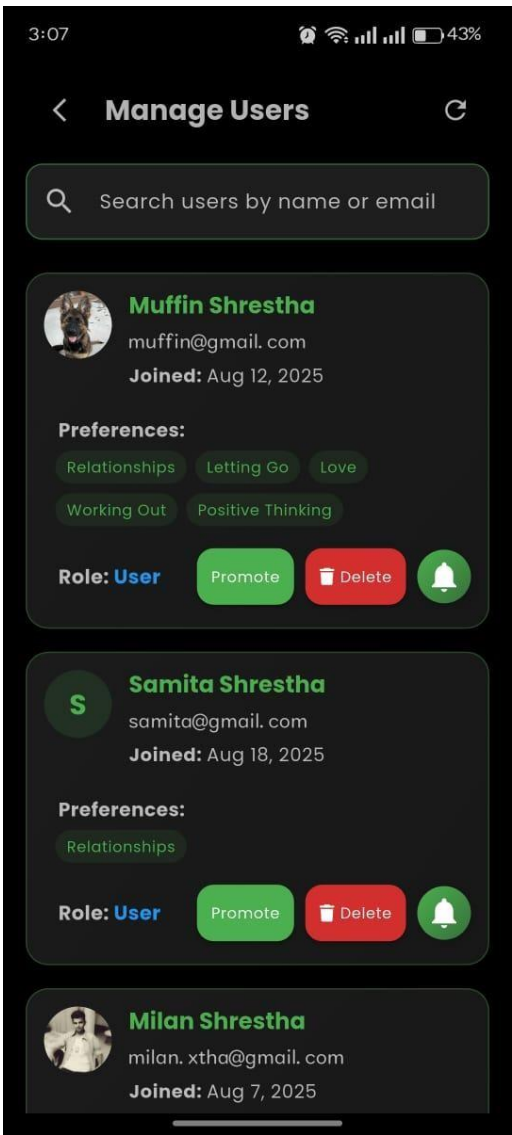
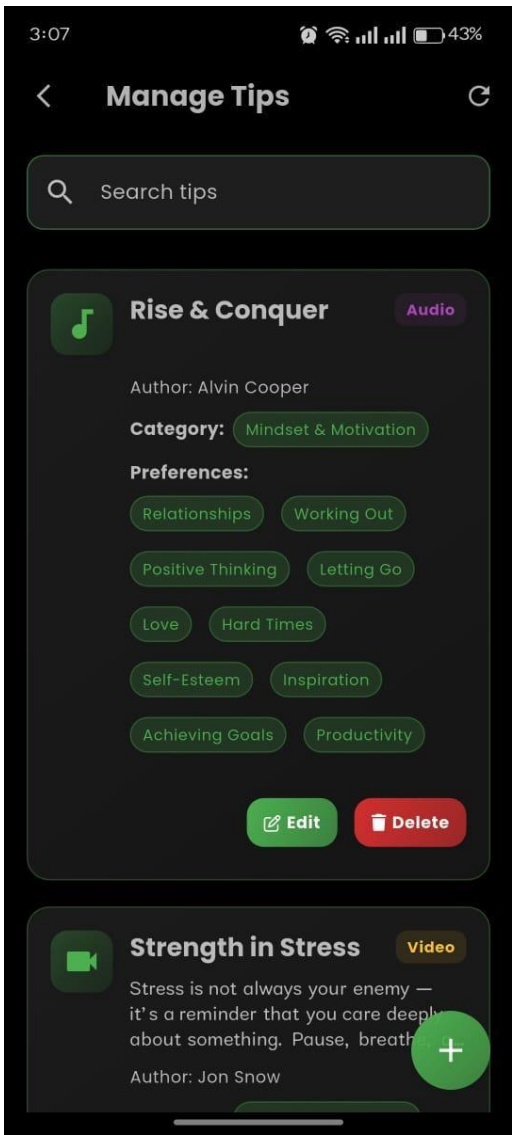


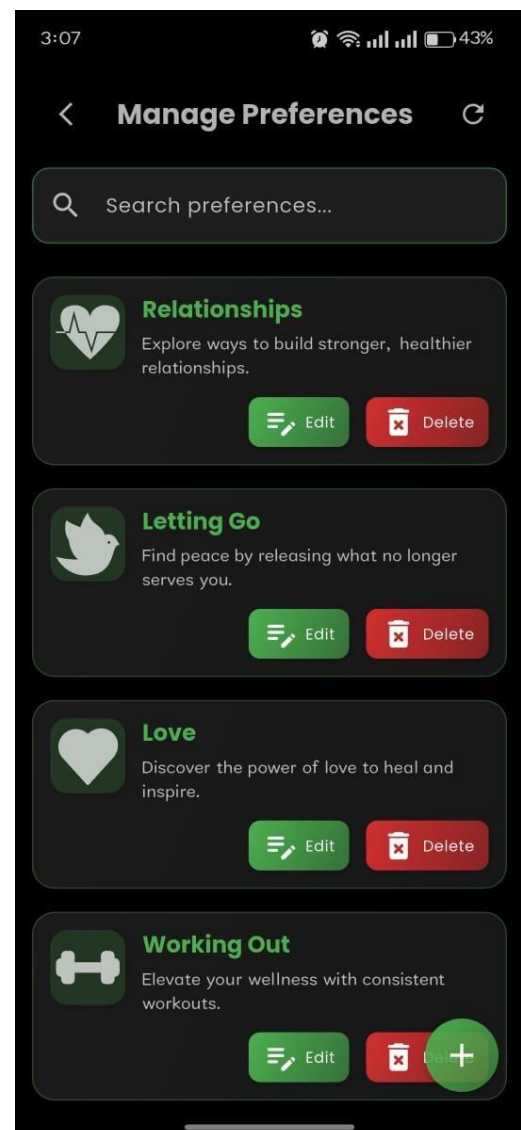
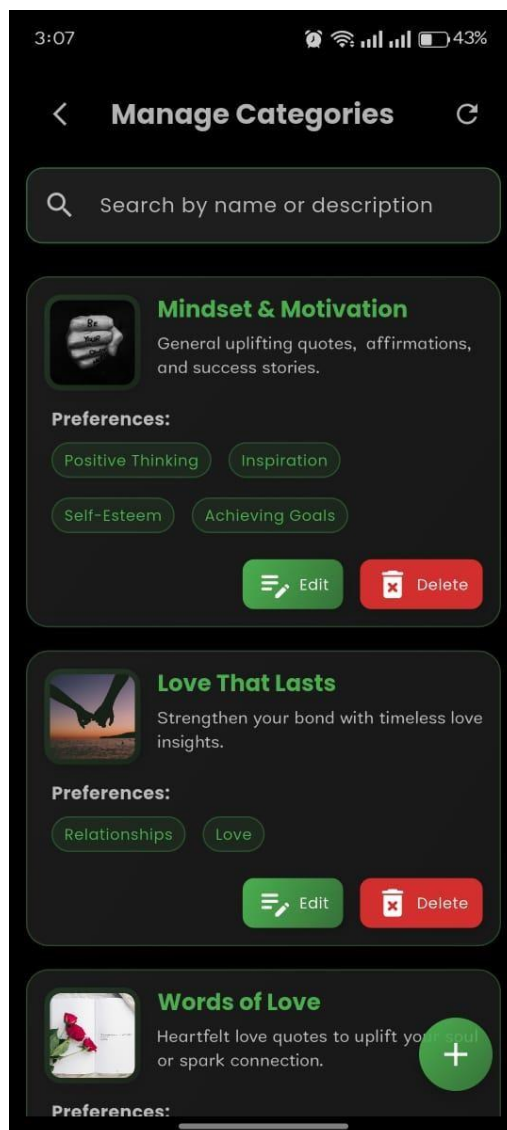
4.5.12. Image Viewer Screen



4.5.13. Admin Dashboard Screen







3:08    43%

< **Add Tip**  **Save**

Tip Type

Quote 

Tip Title

Enter tip title

Tip Description

Enter tip description

Tip Author

Enter author name

Author Icon

 Upload from Device  Enter URL ✓

Enter icon URL (PNG or JPEG)

Category  **Save Tip**

4.6. Development Status with Evidence

Actual coding work has started and the first core pieces of the system are already in place. Development is following the RUP plan: Inception and Elaboration are complete, and Construction Iteration 1 is now underway.

So far, the following parts are implemented and working on a physical Android device and emulator:

Onboarding flow

- Multi-screen introduction explaining Resilio's main features.
- “Get Started” and “Skip” options working as expected.
- Basic preference capture (e.g. language, notification consent) saved locally.

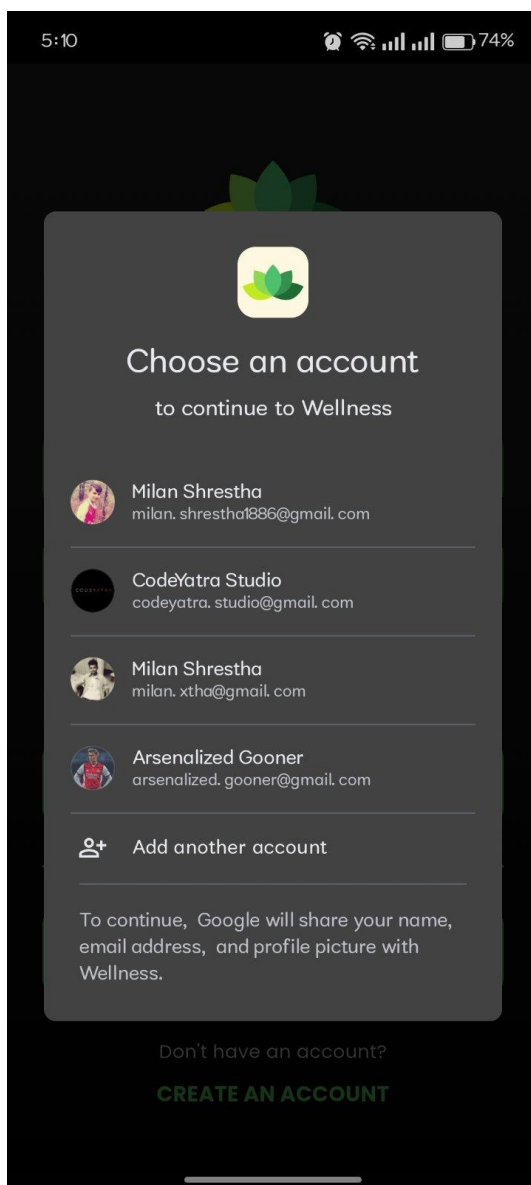
Evidence:

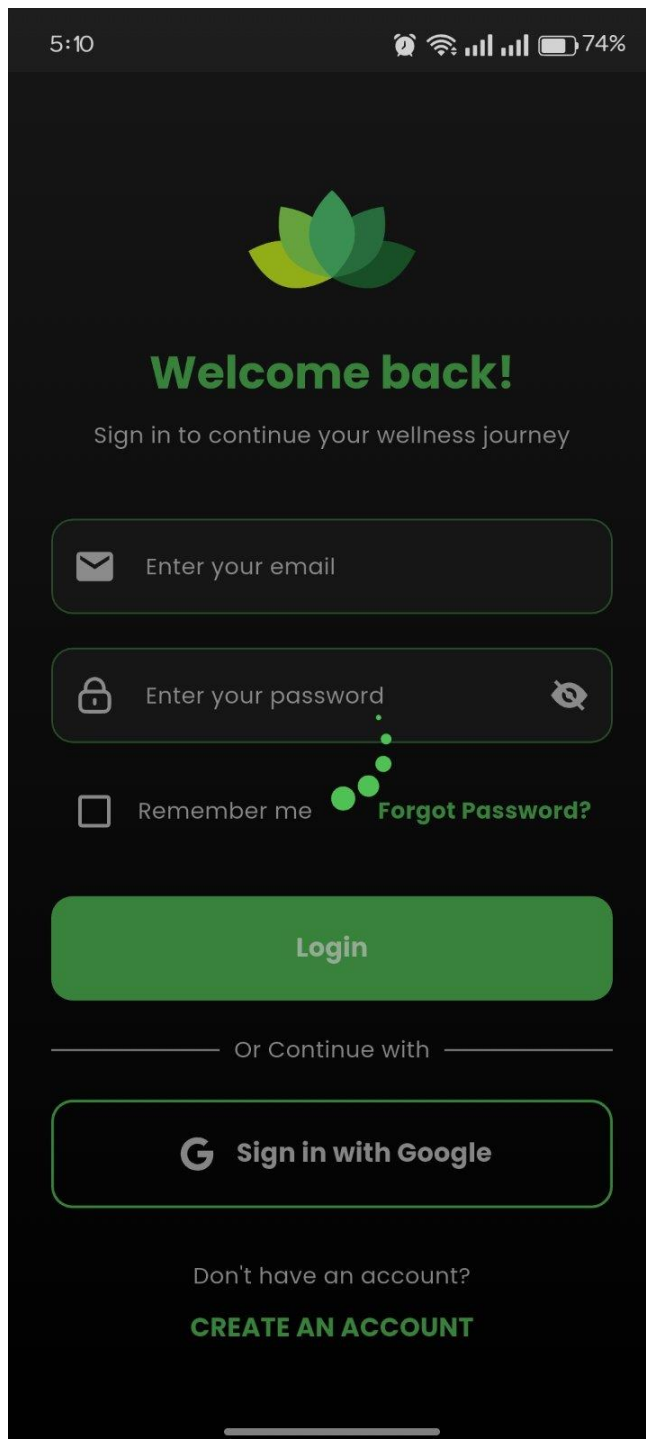


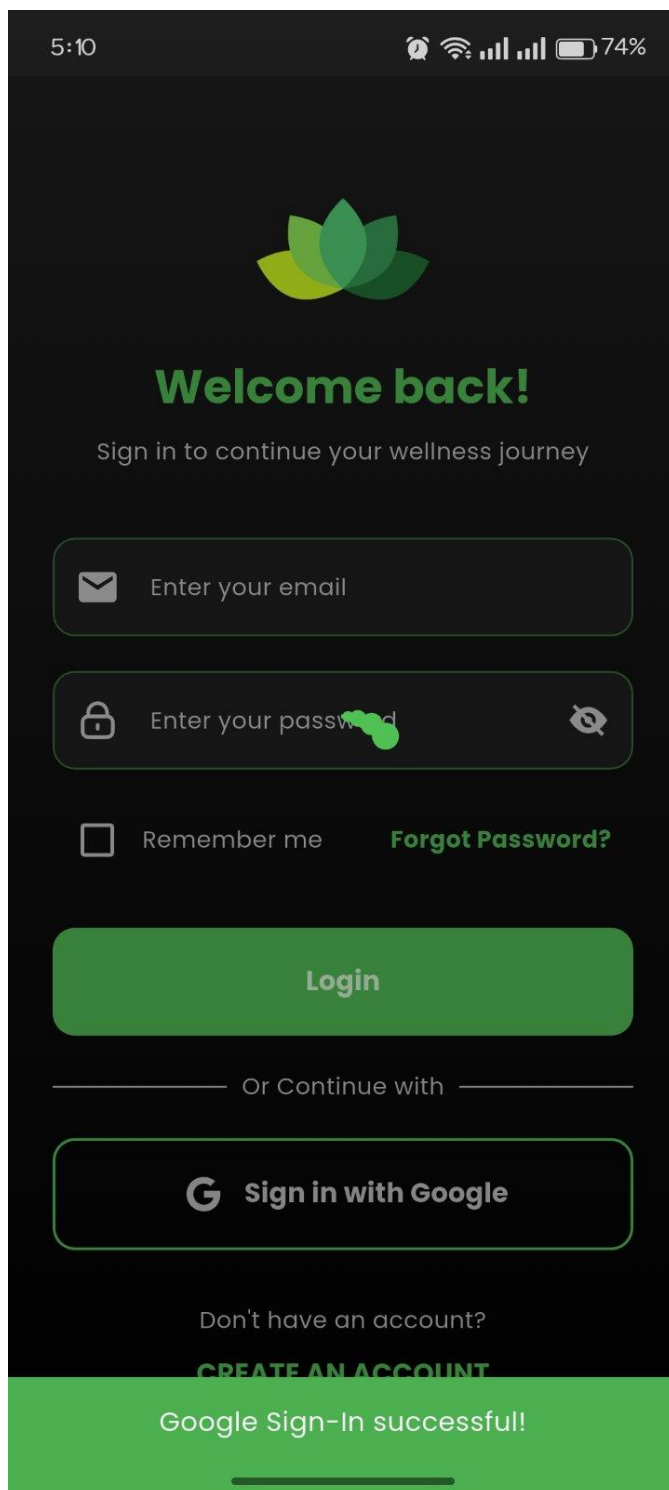
Authentication module

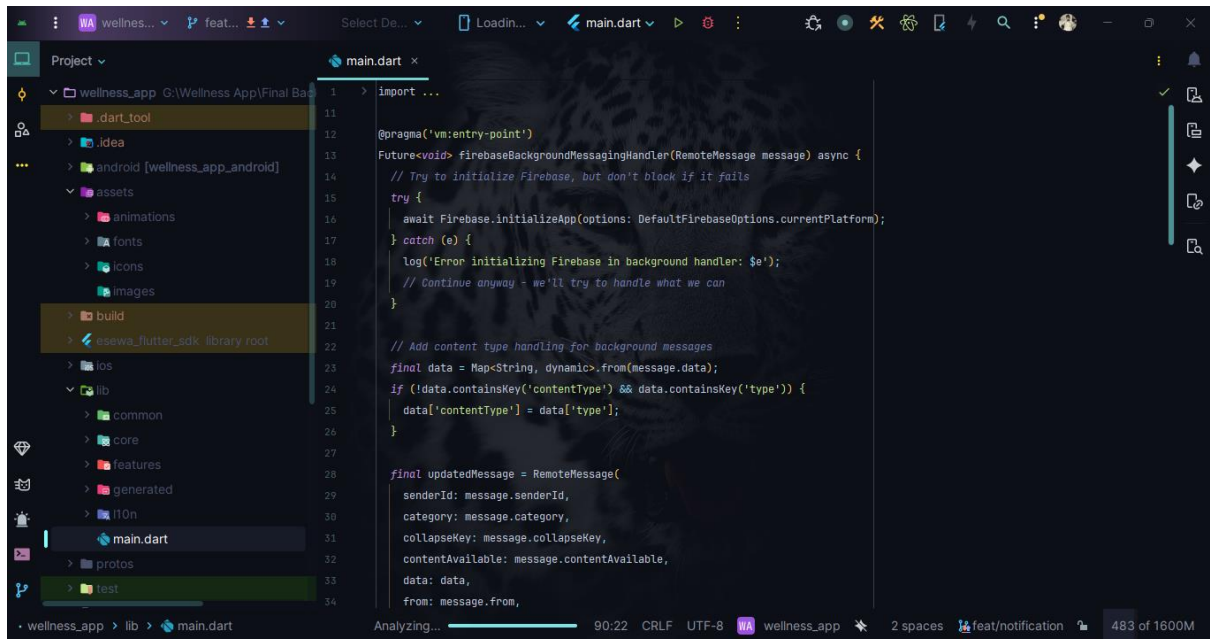
- Email/password registration and login with server-side validation.
- Password reset flow via email.
- Google sign-in fully integrated using Firebase Auth.
- Passwordless login using Supertokens (magic link) tested on Android.
- Authentication state persists between app restarts (JWT stored securely and refreshed).

Evidence:









```
1  import ...
11
12  @pragma('vm:entry-point')
13  Future<void> firebaseBackgroundMessagingHandler(RemoteMessage message) async {
14    // Try to initialize Firebase, but don't block if it fails
15    try {
16      await Firebase.initializeApp(options: DefaultFirebaseOptions.currentPlatform);
17    } catch (e) {
18      log('Error initializing Firebase in background handler: $e');
19      // Continue anyway - we'll try to handle what we can
20    }
21
22    // Add content type handling for background messages
23    final data = Map<String, dynamic>.from(message.data);
24    if (!data.containsKey('contentType') && data.containsKey('type')) {
25      data['contentType'] = data['type'];
26    }
27
28    final updatedMessage = RemoteMessage(
29      senderId: message.senderId,
30      category: message.category,
31      collapseKey: message.collapseKey,
32      contentAvailable: message.contentAvailable,
33      data: data,
34      from: message.from,
```

5. Analysis of Progress

This section reviews development progress against the planned timeline. The project follows Rational Unified Process methodology with defined milestones for each phase.

5.1. Progress Review

- **Inception Phase (November 1-30, 2024) - Complete ✓**

Business modelling defined project vision. Stakeholder interviews revealed specific needs. Competitor analysis examined BetterHelp, Headspace, and Wysa. Risk assessment identified ten potential risks with mitigation strategies. Environment setup configured Flutter, Git, Supabase, Firebase, and Cloudinary.

Milestone achieved: November 30, 2024

- **Elaboration Phase (December 1-31, 2024) - Complete ✓**

System architecture established Flutter frontend, Node.js backend, PostgreSQL server storage, and SQLite offline storage. UI/UX design delivered wireframes and high-fidelity Figma mock-ups. Architectural prototypes validated WebRTC video calling, offline sync, and eSewa payment integration.

Milestone achieved: December 31, 2024

- **Construction Phase (January 1 - March 15, 2025) - In Progress**

Iteration 1 running ahead of schedule. Authentication complete with email, Google, Facebook, and password less login. Onboarding screens and navigation structure complete. Backend API endpoints and database schemas deployed. Core deliverables completed January 8, two days early.

Iteration scheduled for January 1-10.

Core deliverables completed by January 8, two days early.

5.2. Progress Table

Table 1: Progress Table

Phase	Planned	Actual	Status
Business Modelling	Nov 1-8	Nov 1-7	Complete ✓
Initial Requirements	Nov 9-16	Nov 9-15	Complete ✓
Risk Assessment	Nov 17-22	Nov 16-21	Complete ✓
Environment Setup	Nov 23-30	Nov 22-29	Complete ✓
Requirements Refinement	Dec 1-6	Dec 1-6	Complete ✓
System Architecture	Dec 7-13	Dec 7-12	Complete ✓
UI/UX Design	Dec 14-21	Dec 13-20	Complete ✓
Architectural Prototypes	Dec 22-27	Dec 21-27	Complete ✓
Iteration 1: Core Foundation	Jan 1-10	Jan 1-8	Complete ✓
Iteration 2: Content & Multimedia	Jan 11-25	-	Pending
Iteration 3: Offline Support	Jan 26 - Feb 4	-	Pending
Iteration 4: Therapist & Booking	Feb 5-19	-	Pending
Iteration 2: Content & Multimedia	Jan 11-25	-	Pending
Iteration 3: Offline Support	Jan 26 - Feb 4	-	Pending
Iteration 4: Therapist & Booking	Feb 5-19	-	Pending
Iteration 5: Wellness Games	Feb 20 - Mar 1	-	Pending
Iteration 6: Payments & Notifications	Mar 2-9	-	Pending
Iteration 7: Admin & Polish	Mar 10-15	-	Pending

5.3. Action Plan

- **Immediate (January 10-25)**

Complete user profile avatar upload. Begin Iteration 2 focusing on dashboard, content viewers, and media integration.

- **Short-Term (January 26 - February 19)**

Implement offline support in Iteration 3. Build therapist booking with WebRTC in Iteration 4. These are highest-risk components.

- **Stakeholder Communication**

Mid-Construction review with supervisors in early February. Progress update to Ms. Sushana Karki and Dr. Dibyandra Singh after Iteration 4.

6. Future Work

This section outlines remaining tasks following the work breakdown structure timeline. Construction has six more iterations, followed by Transition and Maintenance phases.

6.1. Remaining Construction Iterations

- Iteration 2: Content & Multimedia (January 11-25, 2025)
Personalized dashboard, content viewers with text-to-speech, audio and video players, and Cloudinary media integration.
- Iteration 3: Offline Support (January 26 - February 4, 2025)
SQLite local database, LRU caching, online/offline detection, background sync service, and conflict resolution.
- Iteration 4: Therapist & Booking (February 5-19, 2025)
Multi-step questionnaire, therapist matching algorithm, booking system, WebRTC/Agora video sessions, and in-session chat.
- Iteration 5: Wellness Games (February 20 - March 1, 2025)
Breathing exercise game, wellness trivia with leader board, affirmation builder, and role-playing journey with XP system.
- Iteration 6: Payments & Notifications (March 2-9, 2025)
ESewa SDK integration, subscription packages, Firebase push notifications, and user settings.
- Iteration 7: Admin & Polish (March 10-15, 2025)
Mobile admin app, Flutter Web dashboard, performance optimization.

Milestone: Construction Complete - March 15, 2025

6.2. Transition Phase (March 16 - April 15, 2025)

Quality assurance including performance, security, and compatibility testing (March 16-25). User acceptance testing with mental health professionals and stakeholder sign-off (March 26 - April 2). Android APK build and Desktop dashboard deployment (April 3-9). Documentation and presentation preparation (April 10-15).

6.3. Maintenance Phase (April 16-30, 2025)

Bug tracking, performance monitoring, user support, and KPI tracking for retention metrics.

Milestone: Project Complete - April 30, 2025

6.4. Critical Path

Iteration 3 (offline support) and Iteration 4 (WebRTC sessions) carry highest technical complexity. Two days buffer exists between Construction and Transition.

7. Conclusion

This interim report documents the development progress of Resilio, a mobile mental health application designed specifically for Nepal. The project addresses a critical gap: 92 percent of Nepalis with mental health conditions never receive professional care, yet existing digital solutions are either too expensive, require constant internet, or ignore local cultural context. Resilio integrates therapist booking, live video sessions, mood tracking, questionnaires, wellness games, and motivational content within a single offline-capable platform. Payment through eSewa removes credit card barriers. Nepali language support addresses accessibility concerns that Western apps ignore.

The project follows Rational Unified Process methodology with two of four phases complete. Inception delivered project vision, stakeholder requirements, and environment setup by November 30, 2024. Elaboration produced system architecture, UI/UX designs, and working prototypes validating WebRTC, offline sync, and eSewa integration by December 31, 2024. Construction began January 1, 2025, progressing ahead of schedule. Iteration 1 completed two days early with functional authentication, onboarding screens, and backend API endpoints. Overall completion stands at 38 percent.

Six Construction iterations remain covering content systems, offline support, therapist booking, wellness games, payments, and admin features. Target completion is March 15, 2025, followed by Transition through April 15 and Maintenance through April 30, 2025. The project remains on track with both milestones achieved on schedule. By April 30, 2025, Resilio will offer something that does not currently exist in Nepal: an integrated, affordable, offline-capable mental health platform built for local realities.

8. References

- Arjadi, R., Nauta, M. H., Chowdhary, N. and Bockting, C. L. (2015) 'A systematic review of online interventions for mental health in low- and middle-income countries: a neglected field', *Global Mental Health*, 2, p. e12.
- Backhaus, A., Agha, Z., Maglione, M. L., Repp, A., Ross, B., Zuest, D., Rice-Thorp, N. M., Lohr, J. and Thorp, S. R. (2012) 'Videoconferencing psychotherapy: a systematic review', *Psychological Services*, 9(2), pp. 111-131.
- Bakker, D., Kazantzis, N., Rickwood, D. and Rickard, N. (2016) 'Mental health smartphone apps: review and evidence-based recommendations for future developments', *JMIR Mental Health*, 3(1), p. e7.
- BetterHelp (2023) About BetterHelp. Available at: <https://www.betterhelp.com/about/> (Accessed: 15 December 2024).
- Chandrashekar, P. (2018) 'Do mental health mobile apps work: evidence and recommendations for designing high-efficacy mental health mobile apps', *mHealth*, 4, p. 6.
- Economides, M., Martman, J., Bell, M. J. and Sanderson, B. (2018) 'Improvements in stress, affect, and irritability following brief use of a mindfulness-based smartphone app: a randomized controlled trial', *Mindfulness*, 9(5), pp. 1584-1593.
- Fleming, T. M., de Beurs, D., Khazaal, Y., Gaggioli, A., Riva, G., Botella, C., Baños, R. M., Aschieri, F., Bavin, L. M., Kleiboer, A., Merry, S., Lau, H. M. and Riper, H. (2019) 'Maximizing the impact of e-therapy and serious gaming: time for a paradigm shift', *Frontiers in Psychiatry*, 10, p. 65.
- Headspace (2023) Headspace for Organizations: About Us. Available at: <https://www.headspace.com/organizations/about> (Accessed: 15 December 2024).
- Imel, Z. E., Caperton, D. D., Tanana, M., Atkins, D. C. and Tibbo, M. (2020) 'Annual research review: Using technology to advance research in child and adolescent mental health problems', *Journal of Child Psychology and Psychiatry*, 61(3), pp. 356-366.
- Inkster, B., Sarda, S. and Subramanian, V. (2018) 'An empathy-driven, conversational artificial intelligence agent (Wysa) for digital mental well-being: real-world data evaluation mixed-methods study', *JMIR mHealth and uHealth*, 6(11), p. e12106.
- Kohrt, B. A. and Harper, I. (2008) 'Navigating diagnoses: understanding mind-body relations, mental health, and stigma in Nepal', *Culture, Medicine and Psychiatry*, 32(4), pp. 462-491.
- Luitel, N. P., Jordans, M. J., Adhikari, A., Upadhaya, N., Hanlon, C., Lund, C. and Komproe, I. H. (2015) 'Mental health care in Nepal: current situation and challenges for development of a district mental health care plan', *Conflict and Health*, 9(3), pp. 1-11.
- Ministry of Health and Population (2020) Nepal Mental Health Survey 2020. Kathmandu: Government of Nepal.
- Naeem, F., Sajid, S., Fatima, B., Rana, M., Mushahid, M., Arshad, M. R. and Munshi, T. (2021) 'Digital psychiatry in low- and middle-income countries post-COVID-19: opportunities, challenges, and solutions', *Indian Journal of Psychiatry*, 63(Suppl 3), pp. S376-S379.
- Naslund, J. A., Aschbrenner, K. A., Araya, R., Marsch, L. A., Unützer, J., Patel, V. and Bartels, S. J. (2017) 'Digital technology for treating and preventing mental

disorders in low-income and middle-income countries', *The Lancet Psychiatry*, 4(6), pp. 486-500.

Nepal Telecommunications Authority (2022) MIS Report, 137.

Available at: <https://nta.gov.np/mis-reports>

(Accessed: 15 May 2024).

Patel, V., Chowdhary, N., Rahman, A. and Verdeli, H. (2011) 'Improving access to psychological treatments: lessons from developing countries', *Behaviour Research and Therapy*, 49(9), pp. 523-528.

Patel, V., Saxena, S., Lund, C., Thornicroft, G., Baingana, F., Bolton, P., Chisholm, D., Collins, P. Y., Cooper, J. L., Eaton, J., Herrman, H., Herzallah, M. M., Huang, Y., Jordans, M. J., Kleinman, A., Medina-Mora, M. E., Morgan, E., Niaz, U., Omigbodun, O., Prince, M., Rahman, A., Saraceno, B., Sarkar, B. K., De Silva, M., Singh, I., Stein, D. J., Sunkel, C. and Unützer, J. (2018) 'The Lancet Commission on global mental health and sustainable development', *The Lancet*, 392(10157), pp. 1553-1598.

Shiffman, S., Stone, A. A. and Hufford, M. R. (2008) 'Ecological momentary assessment', *Annual Review of Clinical Psychology*, 4, pp. 1-32.

Torous, J., Andersson, G., Bertagnoli, A., Christensen, H., Cuijpers, P., Firth, J., Haim, A., Hsin, H., Hollis, C., Lewis, S., Mohr, D. C., Pratap, A., Roux, S., Sherrill, J. and Arean, P. A. (2019) 'Towards a consensus around standards for smartphone apps and digital mental health', *World Psychiatry*, 18(1), pp. 97-98.

World Health Organization (2022) Mental health and COVID-19: early evidence of the pandemic's impact, Scientific Brief. Geneva: WHO.

9. Appendix

Appendix A: Detailed Mental Health Statistics in Nepal

[Return to Section [1.2. Current Scenario](#)]

The bigger problem is that almost nobody gets treatment. (Luitel et al. 2015) estimated that over 90 percent of people with mental health issues in Nepal never receive any professional care. There are many reasons for this. The country has very few mental health workers, less than one psychiatrist for every 200,000 people according to WHO data (World Health Organization, 2017). Most services are in Kathmandu and other cities while rural communities have almost nothing. On top of that, many people feel ashamed to talk about mental health or see it as a spiritual matter rather than a medical one (Kohrt and Harper, 2008). Cost and travel also stop people from seeking help.

The pandemic made things more urgent. WHO reported a 25 percent global increase in anxiety and depression during 2020, and Nepal saw similar patterns with increased calls to suicide helplines and mental health services (Sigdel et al., 2021). Studies now show that video therapy can be just as effective as meeting in person (Backhaus et al., 2012), which should make digital solutions more viable. But you still need apps that actually work in local contexts. Right now, if someone in rural Nepal wants mental health support through their phone, they have to choose between expensive foreign apps they cannot afford, basic local apps that offer little real help, or trying to piece together different tools that were never meant to work together.

A.1 Prevalence of Mental Disorders

The Nepal Mental Health Survey conducted in 2020 provides the most comprehensive data on mental health conditions in the country. The survey covered all seven provinces and included both urban and rural populations.

Condition	Lifetime Prevalence (%)	12-Month Prevalence (%)
Any Mental Disorder	12.9	5.8
Depression	4.5	2.1
Anxiety Disorders	3.5	1.8
Alcohol Use Disorder	4.0	2.5
Suicidal Ideation	5.9	2.3
Suicide Attempts	1.4	0.5

Source: Ministry of Health and Population (2020)

These numbers translate to approximately 3.5 million Nepali adults who have experienced some form of mental health condition. Depression and anxiety together affect around 2.3 million people. The suicidal ideation rate of 5.9 percent means roughly

1.7 million adults have thought about ending their lives at some point.

A.2 Treatment Gap Analysis

The treatment gap refers to the percentage of people who need mental health care but do not receive it. Nepal has one of the highest treatment gaps in the world.

Indicator	Nepal	Low-Income Countries Average	High-Income Countries Average
Treatment Gap (Overall)	92%	85%	45%
Treatment Gap (Depression)	89%	80%	50%
Treatment Gap (Severe Mental Disorders)	75%	70%	35%

Source: Luitel et al. (2015); Patel et al. (2018)

This means that for every 100 people with mental health problems in Nepal, only 8 receive any form of professional care. The remaining 92 either cope on their own, rely on traditional healers, or suffer without any support.

A.3 Mental Health Workforce

Nepal has a severe shortage of trained mental health professionals. The following table compares Nepal's mental health workforce with WHO recommendations.

Professional Type	Nepal (per 100,000)	WHO Recommended (per 100,000)	Gap
Psychiatrists	0.36	1.0	64% shortage
Psychologists	0.08	1.0	92% shortage
Mental Health Nurses	0.25	2.0	88% shortage
Social Workers	0.05	1.0	95% shortage

Source: World Health Organization (2017)

With approximately 100 psychiatrists serving a population of 30 million, each psychiatrist would need to care for 300,000 people. Most of these professionals are based in Kathmandu, leaving rural areas with almost no coverage.

A.4 Geographic Distribution of Services

Mental health services in Nepal are heavily concentrated in urban areas:

Region	% of Population	% of Mental Health Services
Kathmandu Valley	12%	65%
Other Urban Areas	18%	25%
Rural Areas	17%	10%

Source: Luitel et al. (2015)

This means 70 percent of the population has access to only 10 percent of mental health services. A person in rural Humla or Dolpa may need to travel 2-3 days to reach the nearest psychiatrist.

A.5 Barriers to Treatment

Research has identified several key barriers preventing Nepalis from seeking mental health care:

Barrier	% Reporting This Barrier
Cost of treatment	78%
Distance to services	65%
Stigma and shame	62%
Belief it is not a medical issue	48%
Lack of awareness about services	45%
Preference for traditional healers	38%

Source: Kohrt and Harper (2008); Luitel et al. (2015)

Cost and distance are the top barriers, followed closely by stigma. Many Nepalis still view mental illness as a result of karma, spiritual possession, or personal weakness rather than a treatable medical condition.

[Return to Section 1.2. Current Scenario]

Appendix B: Extended Problem Analysis

[Return to Section [1.3. Problem Statement](#)]

B.1 App Fragmentation Problem

The current mental health app landscape is highly fragmented. Users who want comprehensive support must use multiple disconnected applications:

Scenario 1: User seeking therapy and self-help

- Downloads BetterHelp for therapist access
- Downloads Daylio for mood tracking
- Downloads Calm for meditation
- Downloads Youper for anxiety exercises

Result: 4 apps that do not share data or work together

Scenario 2: User in Nepal seeking affordable support

- Tries BetterHelp but cannot afford \$70/week
- Downloads free meditation app but content is in English
- Finds Nepali mental health website but it is only informational.

Result: No integrated solution available

Bakker et al. (2016) studied 2,000 mental health app users and found:

- Average user downloads 3.2 mental health apps
- 74% stop using apps within 2 weeks
- Main reason for abandonment: “felt disconnected from real care”
- Users who had therapist integration showed 3x higher retention

B.2 Lack of Integration with Professional Care

Most self-help apps operate completely separate from professional mental health services. This creates several problems:

Problem	Impact
Therapists cannot see client mood data	Treatment decisions based on incomplete information
Self-assessments not shared with professionals	Duplicate testing, wasted time
No continuity between sessions	Progress lost; users disengage
Crisis situations not flagged	Delayed intervention in emergencies
Self-help exercises not reinforced	Users unsure if doing exercises correctly

(Patel et al. 2011) argued that low-resource countries especially need integrated models because they cannot afford to waste limited professional time on tasks that apps could support.

B.3 Long-term Engagement Failures

(Torous et al. 2019) reviewed 1,000+ mental health apps and found critical engagement problems:

Metric	Finding
Apps with evidence-based content	Only 3%
Apps designed for long-term use	Only 8%
Apps with therapist integration	Only 5%
Apps working offline	Only 12%
Apps with gamification elements	Only 15%

Most apps are designed for short-term use or one-time downloads. They lack features that bring users back daily or that create meaningful progress over time.

B.4 Connectivity Requirements

Mental health apps typically require constant internet connectivity:

App	Offline Capability
BetterHelp	None - all features require internet
Calm	Partial - some downloaded meditations
Headspace	Partial - limited offline content
Wysa	Minimal - basic chatbot only
Woebot	None - requires constant connection

For Nepal where 30% of the population lacks regular internet and many areas have unstable connectivity, these requirements make most apps unusable for rural populations.

B.5 User Journey Without Integrated Solution

A typical user journey in Nepal currently looks like:

1. **Awareness Stage:** Person realizes they might need help
2. **Search Stage:** Looks for apps, finds mostly English options
3. **Trial Stage:** Downloads 2-3 free apps
4. **Frustration Stage:** Apps feel disconnected, content irrelevant
5. **Abandonment Stage:** Stops using apps within 2 weeks
6. **Gap Stage:** Returns to no support or traditional healers
7. **Deterioration Stage:** Condition may worsen without intervention

An integrated solution like Resilio aims to break this cycle by providing everything in one place.

[Return to Section 1.3. Problem Statement]

Appendix C: Detailed App Comparison

[Return to Section [1.3. Current Scenario](#)]

C.1 International Apps Analysis

BetterHelp:

Feature	Details
Founded	2013, USA
Primary Service	Online therapy with licensed counsellors
Cost	\$60-90 per week (\$240-360/month)
Languages	English only
Therapist Location	Mostly US, UK, Canada
Offline Mode	Not available
Payment Methods	Credit card, PayPal only
Nepal Accessibility	Very low - cost prohibitive, no local therapists

Headspace:

Feature	Details
Founded	2010, UK/USA
Primary Service	Meditation, mindfulness courses
Cost	\$69.99/year or \$12.99/month
Languages	English, some dubbed content
Professional Support	None - self-help only
Offline Mode	Limited downloaded sessions
Payment Methods	Credit card, mobile payments
Nepal Accessibility	Low - similar barriers as Calm

C.2 Local and Regional Apps

Available Nepali Mental Health Resources:

Resource	Type	Limitations
CMC Nepal Helpline	Phone hotline	Not an app, limited hours
TPO Nepal Resources	Website/PDF	Informational only, not interactive
Various meditation apps	Mobile apps	Basic features, no therapist integration
Facebook support groups	Social media	Privacy concerns, no professional oversight

Most local options are either phone-based hotlines or static informational websites. Interactive, integrated mobile apps designed for Nepal remain largely unavailable.

Appendix D: Detailed Solution Features

[Return to Section [1.4. Project as a Solution](#)]

D.1 Core Feature Breakdown

Therapist Booking System:

Feature	Description
Therapist Profiles	Detailed profiles with qualifications, specializations, languages
Availability Calendar	Real-time availability with time zone support
Session Types	Video call, audio call, chat-based options
Booking Management	Reschedule, cancel, receive reminders
Therapist Matching	Recommendations based on user needs and preferences

Live Session Capabilities:

Feature	Description
Video Sessions	HD video with adaptive quality for low bandwidth
Audio Sessions	High-quality audio with minimal data usage
Session Recording	Optional recording for user review (with consent)
Notes Integration	Therapist can access user mood data during session
Crisis Protocol	Emergency procedures if user shows distress signs

Mood Tracking System:

Feature	Description
Daily Check-ins	Quick mood logging with emoji and scale options
Mood History	Visual graphs showing patterns over time
Trigger Tracking	Log events, activities, or thoughts affecting mood
Insights	AI-generated observations about mood patterns
Therapist Sharing	Option to share mood data with assigned therapist

Mental Health Questionnaires:

Questionnaire	Purpose	Frequency
PHQ-9	Depression screening	Weekly/Monthly
GAD-7	Anxiety screening	Weekly/Monthly
WHO-5	Wellbeing index	Weekly
Custom Assessments	Specific concerns	As needed

All questionnaires are validated instruments adapted for Nepali context.

D.2 Engagement Features

Gamified Exercises:

Game	Mental Health Benefit	How It Works
Breathing Bubble	Anxiety reduction, relaxation	Follow expanding/contracting bubble to regulate breath
Wellness Trivia	Mental health education	Answer questions about mental health facts
Affirmation Builder	Self-esteem, positive thinking	Create and customize personal affirmations

Research shows gamification increases app retention by 40-60% in mental health contexts (Fleming et al., 2019).

Motivational Contents:

Content Type	Frequency	Personalization
Daily Quotes	Every morning	Based on current mood and goals
Wellness Tips	2-3 per day	Matched to user's focus areas
Short Videos	Updated weekly	Local therapists, Nepali context
Podcasts	Updated weekly	Mental health topics in Nepal
Success Stories	Updated monthly	Real recovery journeys (anonymized)

Research shows gamification increases app retention by 40-60% in mental health contexts (Fleming et al., 2019).

Progress and Rewards:

Feature	Description
Streak Tracking	Consecutive days of app usage
Achievement Badges	Milestones for exercises, sessions, mood logs
Progress Reports	Weekly/monthly summaries of improvement
Goal Setting	Set and track personal mental health goals

D.3 Offline Functionality**How Offline Mode Works:**

Component	Offline Capability
Mood Tracking	Full functionality, syncs when online
Saved Exercises	All downloaded content accessible
Questionnaires	Can complete offline, submits when connected
Motivational Content	Cached quotes and tips available
Session Notes	View past session summaries
Progress Data	Full access to historical data

Technical Implementation:

Technology	Purpose
SQLite Database	Local storage of user data
LRU Caching	Efficient content storage
Protocol Buffers	Compressed data for fast sync
Background Sync	Automatic upload when connected
Conflict Resolution	Handles data conflicts gracefully

Data Usage Optimization:

Activity	Purpose
Daily sync	~2-5 MB
Video session (30 min)	~150-300 MB
Audio session (30 min)	~30-50 MB
Content download	~10-20 MB weekly

Users can pre-download content on WIFI and use app features on minimal data.

D.4 Payment Integration**eSewa Integration:**

Feature	Description
Direct Payment	Pay from eSewa wallet
Subscription	Monthly/yearly plans
Session Payment	Pay per therapy session
Content Unlock	Purchase premium content
Refunds	Processed through eSewa

D.5 Cultural Adaptation

Nepali Context Features:

Feature	Description
Language	Full Nepali language support
Cultural Content	Quotes from Nepali literature, local wisdom
Local Therapists	Nepali-speaking, culturally trained professionals
Stigma-Aware Design	Private, discreet app appearance
Family Dynamics	Content addressing family-related stress

Addressing Local Barriers:

Barrier	How Resilio Addresses It
Cost	Affordable pricing, free tier available
Distance	Remote therapy, no travel needed
Stigma	Private app, discreet notifications
Language	Full Nepali support
Internet	Offline functionality
Trust	Local therapists, familiar payment methods

D.6 User Journey with Resilio

Week 1: Onboarding

- Download app, create account
- Complete initial assessment questionnaires
- Set mental health goals
- Explore free content and exercises
- Start daily mood tracking

Week 2-4: Building Habits

- Daily mood logs become routine
- Complete gamified exercises
- Read daily motivational content
- Patterns begin emerging in mood data
- Consider booking first therapy session

Month 2: Integrated Care

- Regular therapy sessions (if chosen)
- Therapist reviews mood data before sessions
- Personalized recommendations based on progress
- Engagement with community features
- Noticeable improvement in tracking

Ongoing: Continuous Support

- Maintenance therapy as needed
- Self-monitoring continues
- Access to new content and features
- Crisis support available if needed
- Long-term mental wellness management

[Return to Section [1.4. Project as a Solution](#)]

Appendix E: Methodology Details

[Return to Section [5. Methodology](#)]

E.1 Detailed SDLC Model Comparison

Agile Model Analysis

Agile emerged from the software industry's frustration with heavyweight processes that could not adapt to changing requirements. The Agile Manifesto, published in 2001, prioritized individuals over processes, working software over documentation, customer collaboration over contracts, and responding to change over following plans (Beck et al., 2001).

Strengths for Resilio:

Strength	Application
Rapid feedback	Users could test mood tracking early
Flexibility	Requirements could evolve based on therapist input
Continuous delivery	Features released incrementally
Team collaboration	Close coordination between developer and client

Weakness for Resilio:

Weakness	Impact
Light documentation	Academic requirements need detailed reports
Scope creep risk	Features could expand beyond timeline
Sprint alignment	Two-week sprints may not match submission dates
Product owner need	Requires dedicated decision-maker unavailable in academic context

Verdict: Agile's flexibility is attractive but its informal documentation and open-ended nature create risks for academic projects with fixed deliverables.

Waterfall Model Analysis

Though not selected for final comparison, Waterfall deserves mention as the traditional alternative. Royce (1970) originally described this sequential approach where each phase completes before the next begins.

Strengths for Resilio:

Strength	Application
Clear phases	Easy progress tracking
Heavy documentation	Meets academic requirements
Predictable timeline	Fixed deadlines manageable
Simple management	Linear progression easy to supervise

Weakness for Resilio:

Weakness	Impact
No iteration	Cannot refine features based on feedback
Late testing	Problems discovered only at end
Rigid requirements	Cannot adapt to user insights
High risk	Major issues surface late in project

Verdict: Waterfall's rigidity makes it unsuitable for user-centred mental health applications where feedback shapes design.

Prototype Model Analysis

Prototyping focuses on building quick versions for user evaluation. Sommerville (2016) describes two approaches: throwaway prototyping where prototypes are discarded after learning, and evolutionary prototyping where prototypes gradually become the final system.

Strengths for Resilio:

Strength	Application
User feedback	Test mood tracking interfaces early
Requirement clarity	Users show what they want, not just describe it
Risk reduction	Identify usability issues before full build
Stakeholder buy-in	Tangible demonstrations build confidence

Weakness for Resilio:

Weakness	Impact
Scope uncertainty	Endless refinement cycles possible
Documentation gaps	Focus on prototypes over reports
Effort duplication	May rebuild features multiple times
No clear phases	Hard to measure progress against deadlines

Verdict: Prototyping offers valuable user feedback but lacks structure for managing complete academic projects.

RUP Model Analysis

(Kruchten, 2003) developed RUP to combine the benefits of iterative development with the predictability of phased approaches. The model uses four phases containing multiple iterations.

Strengths for Resilio:

Strength	Application
Structured iteration	Refine features within defined phases
Use-case driven	Features trace to real user needs
Risk management	Address technical challenges early
Balanced documentation	Sufficient for academic needs
Stakeholder checkpoints	Regular expert review built in

Weakness for Resilio:

Weakness	Impact
Complexity	More overhead than pure Agile
Learning curve	Understanding RUP concept can be tough
Artifact heavy	Many documents to maintain

Verdict: RUP provides the structure academic projects need while allowing iteration that user-centred design requires.

Comparison Matrix

Criteria	Agile	Waterfall	Prototype	RUP
Flexibility	High	Low	High	Medium
Documentation	Light	Heavy	Light	Balanced
User Involvement	Continuous	End Only	Continuous	Phased
Risk Management	Reactive	Late	Reactive	Proactive
Structure	Low	High	Low	Medium

Iteration Support	High	None	High	Medium
Deadline Alignment	Difficult	Easy	Difficult	Easy
Academic Fit	Poor	Medium	Poor	Good

[Return to Section 5.1. SDLC Models Comparison]

Appendix F: Client Letter

Figure 27: Client approval letter from Ms. Sushana Karki

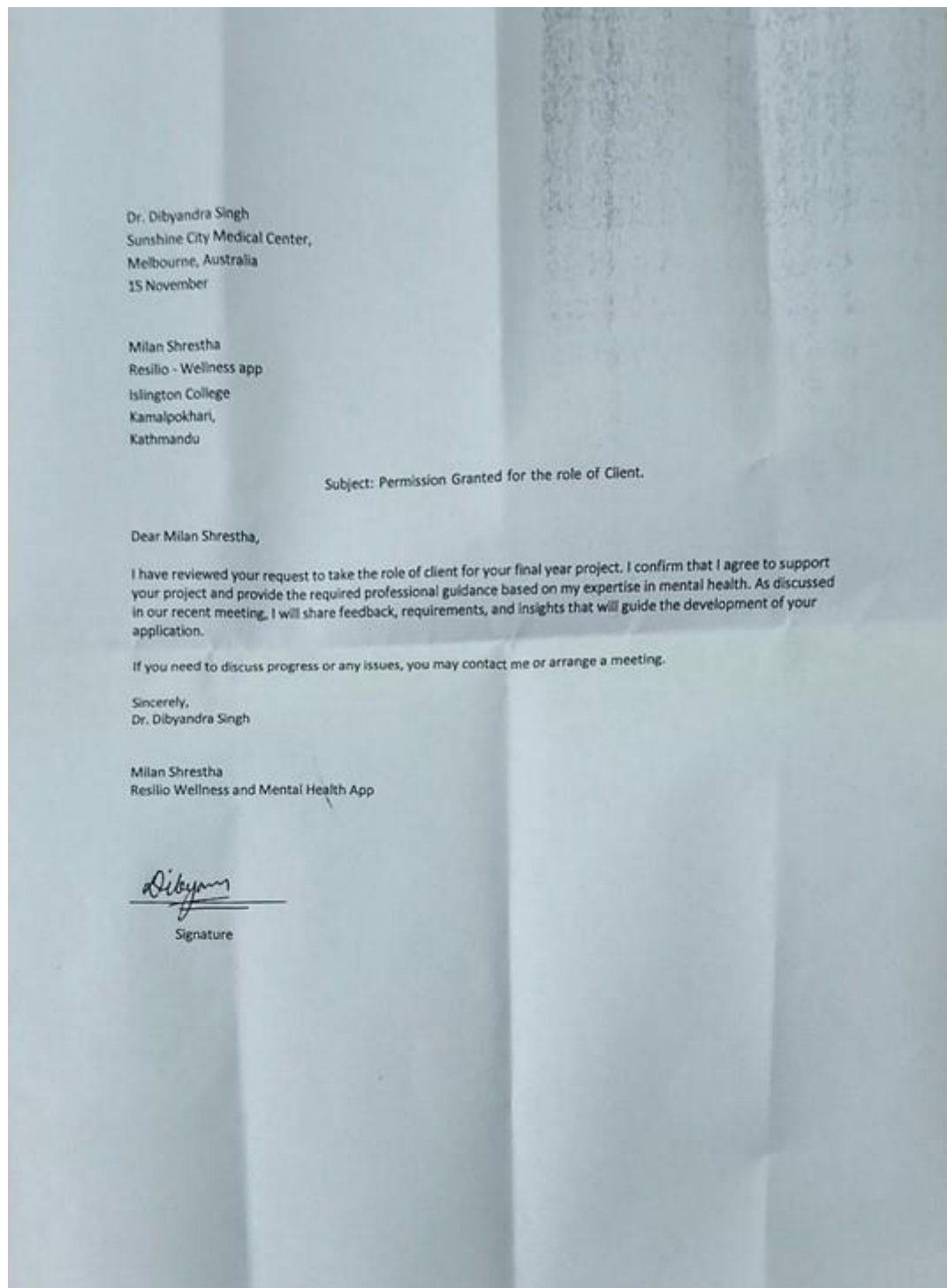


Figure 28: Client approval letter from Dr. Dibyandra Singh

Appendix F: Software Requirement Specification (SRS)

1. Introduction

1.1. Purpose

This Software Requirement Specification document defines the functional and non-functional requirements for Resilio, a mobile mental health application designed for Nepal. The document serves as the primary reference for development, testing, and stakeholder communication throughout the project lifecycle.

1.2. Intended Audience

This document is intended for multiple stakeholders involved in the development and evaluation of Resilio. Developers should focus on system overview, functional requirements, and non-functional requirements to understand implementation specifications. Testers should review functional requirements for designing test cases, performance benchmarks, and security testing criteria. Project supervisors and academic evaluators should read the introduction for project context, overall system description for scope understanding, and functional requirements for feature verification. Mental health consultants and clinical advisors should focus on system features, user classifications, and clinical functionality requirements to ensure therapeutic appropriateness and safety standards are met.

1.3. Scope

In Scope:

- User authentication and profile management
- Therapist booking and live video/audio sessions
- Mood tracking and mental health questionnaires
- Wellness games and motivational content
- Offline functionality and data synchronization
- eSewa payment integration
- Admin dashboard for content management

Out of Scope:

- iOS App Store deployment (Android only for initial release)
- AI-powered chatbot functionality
- Integration with hospital management systems
- Prescription or medication management

1.4. Definitions, Acronyms and Abbreviations

SN	Term	Definition
1	PHQ-9	Patient Health Questionnaire - 9 item depression screening tool
2	GAD-7	Generalized Anxiety Disorder 7-item scale
3	WHO-5	World Health Organization Five Well-Being Index
4	WebRTC	Web Real-Time Communication protocol for video/audio
5	EMA	Ecological Momentary Assessment for mood tracking
6	LRU	Least Recently Used caching algorithm
7	eSewa	Nepal's leading mobile payment wallet
8	WCAG	Web Content Accessibility Guidelines
9	JWT	JSON Web Token for authentication
10	API	Application Programming Interface
11	SDK	Software Development Kit
12	SQLite	Lightweight local database for offline storage
13	OAuth	Open Authorization protocol for secure login
14	HTTPS	Hypertext Transfer Protocol Secure
15	AES-256	Advanced Encryption Standard with 256-bit key

1.5. References

The system utilizes the following technologies and third-party services:

- Flutter SDK 3.x for cross-platform mobile development
- Node.js v18+ with Express for backend API
- PostgreSQL 15 via Supabase for primary database
- SQLite for local offline storage
- Firebase Authentication for user login services
- Agora SDK for WebRTC video and audio calling
- eSewa SDK for payment processing in Nepal
- Cloudinary for media storage and delivery
- Firebase Cloud Messaging for push notifications

1.6. Overview

This document is organized into seven main sections. The introduction establishes purpose, scope, and definitions. Overall description provides system context, features, user classes, and operating environment. Design constraints outline technical limitations. Assumptions and dependencies list external factors affecting the system. Functional requirements detail specific system behaviours. Non-functional requirements specify performance, usability, and reliability standards. Safety requirements address security and data protection measures.

2. Overall Description

2.1. System Perspective

Resilio operates as a standalone mobile application with cloud backend services. The mobile client built with Flutter communicates with a Node.js backend API hosted on Vercel. PostgreSQL via Supabase serves as the primary database. SQLite enables offline data storage on user devices. The system integrates with external services including Firebase for authentication, eSewa for payments, Cloudinary for media storage, and Agora for real-time video and audio communication.

2.2. System Features

The system provides twelve core features:

- SF1 Authentication: Email, Google, Facebook, and passwordless login options
- SF2 User Profile: Profile management, preferences, and avatar customization
- SF3 Mood Tracking: Daily mood logs with triggers, patterns, and visualizations
- SF4 Questionnaires: PHQ-9, GAD-7, and WHO-5 validated assessments
- SF5 Therapist Booking: Browse profiles, view availability, and schedule sessions
- SF6 Live Sessions: WebRTC video and audio with in-session chat
- SF7 Wellness Content: Motivational quotes, tips, videos, and audio relaxation
- SF8 Games: Breathing exercises, trivia, affirmation builder, and role-playing journey
- SF9 Offline Mode: Local storage with automatic background synchronization
- SF10 Payments: eSewa integration for subscriptions and session payments
- SF11 Notifications: Push and local reminders for sessions and activities
- SF12 Admin Dashboard: Content, user, and therapist management interface

2.3. User Classes and Characteristics

UC1: End User (Patient)

Primary users are Nepali adults aged 18 to 45 seeking mental health support. They possess basic smartphone skills and access the system through the Android mobile app. Key actions include mood logging, completing questionnaires, booking therapy sessions, accessing wellness content, and playing therapeutic games.

UC2: Therapist

Secondary users are licensed mental health professionals registered with Nepal Psychological Association or Nepal Psychiatric Association. They have moderate digital literacy and access the system through the mobile app with elevated privileges. Key actions include conducting live sessions, viewing client mood data and questionnaire results, managing availability schedules, and delivering therapeutic content

UC3: Administrator

Tertiary users are platform managers with advanced digital literacy. They access the system through both web dashboard and mobile admin app. Key actions include content management, therapist credential verification, user support, and analytics monitoring.

2.4. Operating Environment

The mobile application requires Android 8.0 (API 26) or higher with minimum 4GB RAM and 100MB available storage. Internet connection is required for synchronization and live therapy sessions, though core features function offline. The web dashboard is compatible with modern browsers including Chrome, Firefox, and Edge. Backend services are hosted on Vercel with Supabase database hosting. Video sessions require minimum 3G connectivity with automatic fallback to audio-only mode on slower connections.

3. Design and Implementation Constraints

The system operates under the following constraints:

- Internet connection required for live therapy sessions and initial data synchronization
- eSewa account required for accessing paid features and booking therapy sessions
- Therapists must complete verification process before gaining platform access
- Video sessions require minimum 3G network connectivity
- Local offline storage limited to 500MB for cached content
- Single language display per session, either Nepali or English
- Android platform only for initial release phase

4. Assumptions and Dependencies**Assumptions:**

- Users have access to Android smartphones running version 8.0 or higher
- eSewa payment service remains operational and accessible throughout Nepal
- Therapists maintain valid professional licenses with recognized associations

- Users grant necessary device permissions including camera, microphone, storage, and notifications
- Basic digital literacy exists among target user population

Dependencies:

- Firebase Authentication service availability for login functionality
- Agora SDK for WebRTC video and audio session functionality
- Supabase for PostgreSQL database hosting and real-time features
- Cloudinary for media storage and content delivery
- Firebase Cloud Messaging for push notification delivery
- eSewa SDK for payment processing within Nepal

5. Product Functional Requirements

5.1. Authentication Module

ID	Requirement
FR1.1	System shall allow user registration via email and password with validation
FR1.2	System shall support Google OAuth 2.0 login integration
FR1.3	System shall support Facebook OAuth login integration
FR1.4	System shall provide passwordless login via magic link email
FR1.5	System shall implement password reset functionality via email
FR1.6	System shall maintain user sessions using JWT tokens with refresh capability
FR1.7	System shall support secure logout clearing local session data
ID	Requirement

5.2. Mood Tracking Module

ID	Requirement
FR2.1	System shall allow daily mood entry using emoji-based scale (1-5)
FR2.2	System shall capture optional trigger notes with each mood entry
FR2.3	System shall display mood history as visual graph over time
FR2.4	System shall identify and highlight mood patterns automatically
FR2.5	System shall store mood data locally for offline access
FR2.6	System shall sync mood data with server when connectivity returns
FR2.7	System shall share mood data with assigned therapist before sessions

5.3. Questionnaire Module

ID	Requirement
FR3.1	System shall implement PHQ-9 depression screening questionnaire
FR3.2	System shall implement GAD-7 anxiety screening questionnaire
FR3.3	System shall implement WHO-5 wellbeing index questionnaire
FR3.4	System shall calculate and display scores with interpretation
FR3.5	System shall use questionnaire responses for therapist matching algorithm
FR3.6	System shall allow questionnaire completion offline with later sync

5.4. Therapist Booking Module

ID	Requirement
FR4.1	System shall display therapist profiles with qualifications and specializations
FR4.2	System shall show real-time therapist availability calendar
FR4.3	System shall allow session booking with date and time selection
FR4.4	System shall send booking confirmation and reminder notifications
FR4.5	System shall support session rescheduling with 24-hour notice
FR4.6	System shall support session cancellation with refund policy enforcement

5.5. Wellness Content Module

ID	Requirement
FR6.1	System shall deliver daily motivational quotes based on user preferences
FR6.2	System shall provide categorized wellness tips and articles
FR6.3	System shall play audio relaxation content with background playback
FR6.4	System shall stream therapist video content with adaptive quality
FR6.5	System shall cache content locally for offline access
FR6.6	System shall track content consumption for personalization

5.6. Games Module

ID	Requirement
FR7.1	System shall provide breathing exercise game with visual bubble animation
FR7.2	System shall implement wellness trivia with scoring and leaderboard
FR7.3	System shall offer affirmation builder with customization and reminders
FR7.4	System shall include role-playing journey game with XP progression
FR7.5	System shall track streaks and award achievement badges

5.7. Payment Module

ID	Requirement
FR8.1	System shall integrate eSewa SDK for payment processing
FR8.2	System shall offer subscription tiers (Gold, Platinum, Diamond)
FR8.3	System shall process per-session payments for therapy bookings
FR8.4	System shall lock premium content for free-tier users
FR8.5	System shall maintain transaction history for users
FR8.1	System shall integrate eSewa SDK for payment processing

5.8. Notification Module

ID	Requirement
FR10.1	System shall send push notifications for session reminders
FR10.2	System shall send local notifications for mood logging prompts
FR10.3	System shall allow users to configure notification preferences
FR10.4	System shall use discreet notification content for privacy

5.9. Admin Module

ID	Requirement
FR11.1	System shall provide content management interface for quotes, tips, and media
FR11.2	System shall enable therapist credential verification workflow
FR11.3	System shall display user engagement analytics and metrics
FR11.4	System shall implement role-based access control for admin functions

6. Non-Functional Requirements

Performance:

- Application shall launch within 3 seconds on supported devices
- API responses shall complete within 500 milliseconds under normal load
- Video sessions shall maintain stable connection with less than 200ms latency

Availability and Reliability:

- Backend services shall maintain 99.5% uptime availability
- System shall support 10,000 concurrent users without degradation
- Offline mode shall function without data loss for up to 30 days

Usability:

- Interface shall comply with WCAG 2.1 AA accessibility standards
- Application shall support both Nepali and English languages
- New users shall complete onboarding within 5 minutes

Maintainability:

- Codebase shall follow modular architecture with documentation
- System shall support over-the-air updates without app store resubmission
- Logs shall capture errors for debugging and monitoring

7. Safety and Security Requirements

Data Protection:

- All sensitive data shall be encrypted using AES-256 encryption at rest
- All network communication shall use HTTPS with TLS 1.3
- Passwords shall be hashed using bcrypt

Authentication and Authorization:

- User authentication shall implement OAuth 2.0 with JWT tokens
- Sessions shall timeout after 30 minutes of inactivity
- Role-based access control shall restrict therapist access to assigned clients only

Privacy:

- User consent shall be required before sharing data with therapists
- Notification content shall be discreet to protect user privacy
- Data retention shall comply with applicable Nepal regulations

Crisis Protocol:

- System shall implement detection for distress indicators in mood data
- Crisis resources shall be readily accessible within the application
- Emergency contact information shall be available for users in distress

Security Auditing:

- Regular security audits shall be conducted during development
- Vulnerability scanning shall be performed before each release
- Penetration testing shall be completed before production deployment